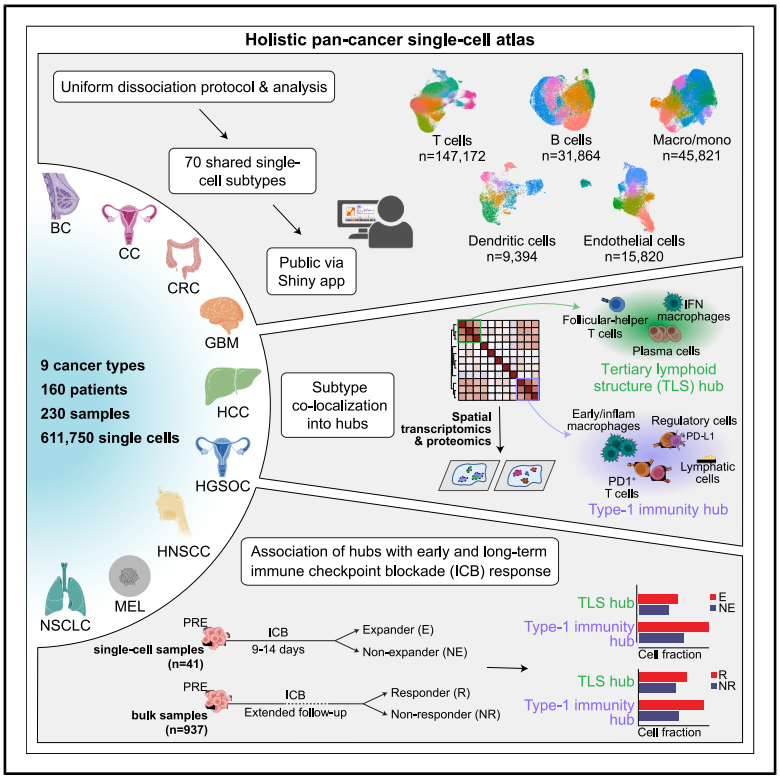
Cell Reports
Medicine

Article

Decoding tumor heterogeneity: A spatially informed pan-cancer analysis of the tumor microenvironment

Graphical abstract



Highlights

- A scRNA-seq atlas across 9 cancer types reveals 70 shared cell subtypes
- Two TME hubs contain co-localized immune reactive cell subtypes
- These hubs and subtypes correlate with early and long-term immunotherapy response
- An interactive Shiny app enables exploration

Authors

Francesca Lodi, Sam Vanmassenhove, Duojiao Chen, ..., Junbin Qian, Jean-Christophe Marine, Diether Lambrechts

Correspondence

diether.lambrechts@kuleuven.be

In brief

Lodi et al. create a pan-cancer single-cell atlas characterizing immune cell heterogeneity within the tumor microenvironment (TME). They identify 70 shared cell subtypes, some of which are spatially co-localized to form two distinct immune reactive TME hubs. Both hubs associate with improved checkpoint immunotherapy outcome across different cancer types.

Lodi et al., 2025, Cell Reports Medicine 6, 102416
October 21, 2025 © 2025 The Authors. Published by Elsevier Inc.
https://doi.org/10.1016/j.xcrm.2025.102416

CellPress

Cell Reports Medicine

CellPress
OPEN ACCESS

Article

Decoding tumor heterogeneity: A spatially informed pan-cancer analysis of the tumor microenvironment

Francesca Lodi,[1,2] Sam Vanmassenhove,[1,2] Duojiao Chen,[3] Bram Boeckx,[1,2] Lucas Ferreira Maciel,[2,4] Frederik Peeters,[2,5] Elena Donders,[2,6,7] Heesoo Song,[1,2] Luuk Harbers,[8] Joanna Poźniak,[2,4] Liselore Loverix,[9,10] Yiduo Zhou,[3] Michel Bila,[11,12] Ayse Bassez,[1,2] Sarah Cappuyns,[5] Tom Venken,[1,2] Siel Olbrecht,[9,10] Amelie Franken,[1,2] Pierre Van Mol,[6,7] Rogier Schepers,[1,2] Thomas van Brussel,[1,2] Gino Philips,[1,2] Hanne Vos,[13] Machteld Keupers,[14] Frederik De Smet,[15,16] Sabine Tejpar,[5] Oliver Bechter,[11,12] Gabriele Bergers,[2,17] Ann Smeets,[13] Paul Clement,[11,12] Els Wauters,[6,7] Jeroen Dekervel,[5] Toon Van Gorp,[9,10] Junbin Qian,[3,18] Jean-Christophe Marine,[2,4] and Diether Lambrechts[1,2,19,*]

[1]Laboratory for Translational Genetics, Department of Human Genetics, KU Leuven, Leuven, Belgium
[2]VIB Center for Cancer Biology, Leuven, Belgium
[3]Zhejiang Key Laboratory of Precision Diagnosis and Therapy for Major Gynecological Diseases, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China
[4]Laboratory for Molecular Cancer Biology, Department of Oncology, KU Leuven, Leuven, Belgium
[5]Digestive Oncology, Department of Gastroenterology, University Hospitals Leuven, Leuven, Belgium
[6]Department of Pneumology, Division of Respiratory Oncology, University Hospitals Leuven, Leuven, Belgium
[7]Laboratory of Respiratory Diseases and Thoracic Surgery (BREATHE), Department of Chronic Diseases and Metabolism, KU Leuven, Leuven, Belgium
[8]Bioinformatics Expertise Center, VIB Center for Cancer Biology, Leuven, Belgium
[9]Department of Obstetrics and Gynaecology, Division of Gynaecological Oncology, University Hospitals Leuven, Leuven, Belgium
[10]Laboratory of Gynaecologic Oncology, Department of Oncology, KU Leuven, Leuven, Belgium
[11]Department of General Medical Oncology, University Hospitals Leuven, KU Leuven, Leuven, Belgium
[12]Laboratory of Experimental Oncology (LEO), Department of Oncology, KU Leuven, Leuven, Belgium
[13]Department of Surgical Oncology, University Hospitals Leuven, KU Leuven, Leuven, Belgium
[14]Department of Radiology, University Hospitals Leuven, KU Leuven, Leuven, Belgium
[15]Laboratory for Precision Cancer Medicine, Translational Cell and Tissue Research Unit, Department of Imaging and Pathology, KU Leuven, Leuven, Belgium
[16]Leuven Institute for Single-cell Omics (LISCO), KU Leuven, Leuven, Belgium
[17]Laboratory of Tumor Microenvironment and Therapeutic Resistance, Department of Oncology, KU Leuven, Leuven, Belgium
[18]Institute of Genetics, Zhejiang University School of Medicine, Hangzhou, China
[19]Lead contact
*Correspondence: diether.lambrechts@kuleuven.be
https://doi.org/10.1016/j.xcrm.2025.102416

SUMMARY

Pan-cancer single-cell atlases explore the heterogeneity of cell types residing within the tumor microenvironment (TME). So far, atlases focused on individual cell types, failing to capture the full complexity of the TME. Here, we present a single-cell atlas that simultaneously considers heterogeneity in 5 cell types, collected from 230 treatment-naive samples across 9 cancer types. We identify 70 pan-cancer single-cell subtypes, investigate their patterns of co-occurrence and show an enrichment of specific subtypes in certain TMEs, e.g., immune-reactive versus immune-suppressive TME. We observe two TME hubs of strongly co-occurring subtypes: one hub resembling tertiary lymphoid structures (TLSs), another consisting of immune-reactive PD1+/PD-L1+ immune-regulatory T cells and B cells, dendritic cells and inflammatory macrophages. Subtypes belonging to each hub are spatially co-localized, while their abundance associates with early and long-term checkpoint immunotherapy response. We publicly share our atlas using a Shiny app, allowing others to explore TME heterogeneity in different biological contexts.

INTRODUCTION

Tumor infiltrating T cells are key functional components of the tumor microenvironment (TME) that have demonstrated anti-cancer efficacy in multiple clinical settings, including immune checkpoint blockade (ICB). Intra-tumoral T cells are phenotypically and functionally heterogeneous cells, displaying various differentiation and functional states. Besides T cells, other tumor-infiltrating immune cells exist. Their heterogeneity and interaction with T cells, as well as their contribution to ICB outcomes, have been insufficiently explored.[1]

Advances in single-cell RNA sequencing (scRNA-seq) technologies have revolutionized our ability to characterize TME heterogeneity. While numerous studies characterized

This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

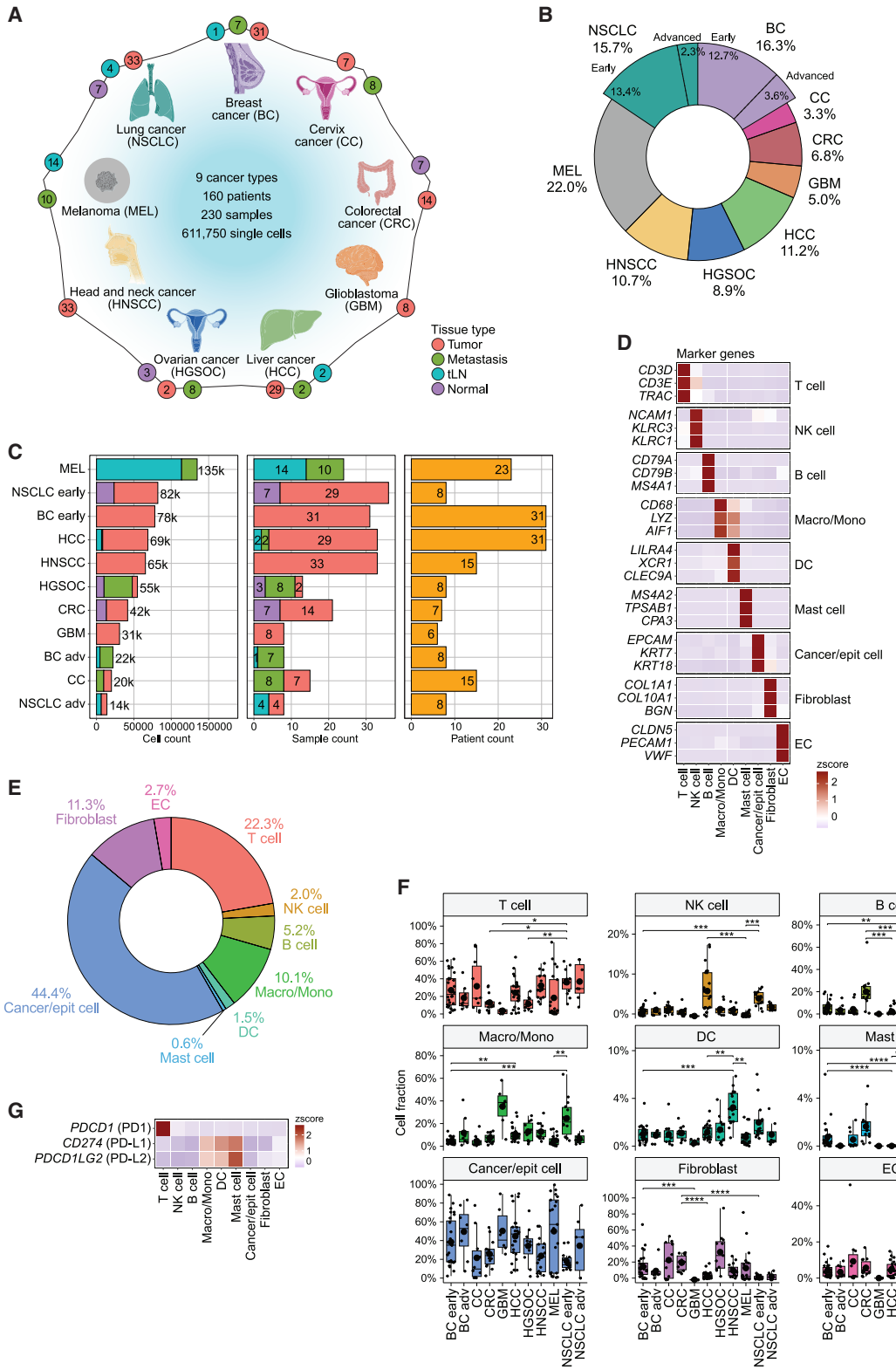


Figure 1. Pan-cancer scRNA-seq analysis of cell types

(A) Schematic overview of the tumor samples included in the study.

(B) Pie chart displaying the percentage of cells obtained by scRNA-seq across cancer types, divided into early and advanced lung and breast cancer.

(legend continued on next page)

CellPress
OPEN ACCESS

intra-tumoral T cells in one or related types of cancer,[2–5] pan-cancer scRNA-seq atlases focusing on T cell heterogeneity have recently been constructed, depicting the various states of T cells and their abundances across cancer types and revealing diverse paths to T cell activation, cytotoxicity, and exhaustion.[6,7] Other pan-cancer atlases have also been constructed, each focusing on the heterogeneity within other cell types, such as intra-tumoral macrophages, natural killer (NK) cells, or cancer-associated fibroblasts.[8–10] Follow-up analyses of the TME at a pan-cancer single-cell level are still highly warranted, because of the following. (1) A limited number of pan-cancer studies is unlikely sufficient to capture all the possible immune states that exist in the TME. (2) While previous pan-cancer studies focused on annotating heterogeneity within an immune cell type, the functional role and clinical significance of various subtypes residing in the TME remain unclear, especially in the context of ICB treatment outcome. (3) Pan-cancer atlases have focused on an individual cell type, often by including studies that only consider single-cell data of the selected cell type. While such approach allows the specific cell type to be studied at greater depth, it fails to study its interaction with other cell types, thereby limiting the potential to comprehensively investigate its role within the TME. (4) Finally, inherent to scRNA-seq technologies, the spatial localization of each cell within the TME is lost. With the increased availability of spatial transcriptomic data, it is becoming possible to also explore these pan-cancer atlases at a spatial level, offering novel possibilities to study the collective behavior of heterogeneous cell subtypes as spatially organized communities. However, pan-cancer studies encompassing all the relevant TME cell types, their relative abundances, interactions and spatial locations are still currently lacking.

Here, by deploying a standardized unbiased protocol for tissue dissociation, we generate a pan-cancer single-cell dataset consisting of 9 different cancer types. We comprehensively characterize the different subtypes of immune cells and based on their within-sample co-occurrences and correlation with T cell reactivity, we identify which subtypes contribute to specific immune-reactive or immune-suppressive TMEs. Spatial characterization across 6 cancer types confirms the co-localization of the subtypes in a spatial context and reveals their organization into distinct hubs. Additionally, we evaluate the clinical relevance of our atlas during early and long-term clinical response to ICB. To facilitate mining of this rich data resource, we build an interactive web portal that can be freely explored.

RESULTS

A pan-cancer scRNA-seq atlas of 9 cancer types

We collected 230 tissue samples from 160 patients diagnosed with one of the following cancer types: breast cancer (BC), cervix carcinoma (CC), colorectal cancer (CRC), glioblastoma multiforme (GBM), head and neck squamous cell carcinoma (HNSCC), hepatocellular carcinoma (HCC), high-grade serous ovarian carcinoma (HGSOC), melanoma (MEL) and non-small cell lung cancer (NSCLC) (Figures 1A–1C). All samples were treatment-naive lesions, except for some CC and HCC samples receiving prior chemotherapy (see STAR Methods). Each cancer type involved samples collected at a specific disease stage: early-stage tumors and non-malignant adjacent tissues were collected during resection, whereas for advanced (metastatic) disease, needle biopsies either from the primary or metastatic tumor or from tumor-invaded lymph nodes (tLNs) were taken (Figures 1A–1C; Table S1). All tissues were immediately digested into a single-cell suspension, the majority (61.3%) was subjected to 5′-scRNA-seq (10× Genomics), although 3′-scRNA-seq was also performed. We obtained expression data for 611,750 high-quality single cells (see STAR Methods) with on average 1,358 genes detected per cell. Among these, 53.9% of samples were derived from primary tumors, while 9.1%, 15.2%, and 21.7% were from tLNs, metastatic tumors, and non-malignant tissue, respectively (Figure 1C).

Pan-cancer single-cell transcriptomic landscape of the TME

We analyzed each cancer type separately to identify cancer and epithelial cells, as well as endothelial cells (ECs), fibroblasts, immune cells, including dendritic cells (DCs), macrophages/monocytes, mast cells, B cells, NK cells, and T cells (Figures 1D and S1A). Additionally, we identified tissue-specific cell types such as oligodendrocytes and microglia in GBM, alveolar cells in NSCLC and hepatocytes in HCC. To estimate the effect of dissociation bias on the proportion of cell types obtained, we subjected samples from 4 different cancer types (NSCLC, HCC, HGSOC, and CRC; $n = 25$) to bulk RNA-seq and compared cell type fractions estimated by deconvolution of the bulk data versus those obtained by scRNA-seq. Immune cells were modestly enriched in the scRNA-seq data, but the enrichment was similar for each of the 4 cancer types (Figure S1B). Since we used the same dissociation protocol across cancer types, this indicates that cell type proportions can reliably be compared between cancer types. All cells belonging to a specific cell type

(C) Number of cells, samples and patients across cancer types and tissue types, color-coded for tissue type as in (A).
(D) Heatmap showing normalized expression of 3 marker genes for each major cell type shared across cancer types. Tissue-specific marker genes were used to delineate cancer/epithelial (cancer/epit) cells.
(E) Pie chart displaying the percentage of cell types detected across cancer types.
(F) Boxplots displaying the percentage of major cell types detected across cancer types. Fractions were calculated for each sample with >500 cells ($n = 160$) (see STAR Methods). Middle line: median; dot: mean; box edges: 25th and 75th percentiles.
(G) Heatmap of normalized *PD1*, *PD-L1*, and *PD-L2* expression in cell types across cancer types.
NK cell, natural killer cell; Macro/mono, macrophage/monocyte; DC, dendritic cell; EC, endothelial cell; tLN, tumor-invaded lymph node; BC, breast cancer; CC, cervical cancer; CRC, colorectal cancer; GBM, glioblastoma multiforme; HCC, hepatocellular carcinoma; HGSOC, high-grade serous carcinoma; HNSCC, head and neck squamous cell carcinoma; MEL, melanoma; NSCLC, non-small cell lung cancer. Non-malignant adjacent tissue was considered as normal tissue. Wilcoxon rank-sum test, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

were therefore merged into one dataset, irrespective of cancer type (Figure 1E). Cancer/epithelial cells were most abundant in MEL, while T cells were most frequent in advanced and early NSCLC. DCs were particularly frequent in HNSCC, whereas macrophages/monocytes were most abundant in early NSCLC and GBM (Figure 1F). *PDCD1* (PD1) was detected exclusively in T cells, while *CD274* and *PDCD1LG2* (PD-L1/PD-L2) were variably expressed by DCs, macrophages or mast cells (Figure 1G).

Then, we subclustered each cell type while applying *Harmony* to correct for potential batch effects between 5′ and 3′ scRNA-seq. Cancer cells and fibroblasts were excluded from this analysis because they were characterized by considerable patient variability. Likewise, due to the low number of cells detected, we did not subcluster mast cells. Following *Harmony*, we calculated Local Inverse Simpson's Index (LISI) scores to confirm the absence of batch effects in the subclusters (Figure S1C). Following the annotation of subclusters based on marker genes, up to 70 cell subtypes were identified. While we briefly describe their main characteristics subsequently, further details about their annotation, marker genes, and abundance in each cancer type can be found in the Shiny app.

Exhausted and regulatory T cell subtypes across cancer types

Based on canonical immune markers and published gene signatures[7] (Table S2), we identified 2 NK cell and 16 T cell subtypes. The latter were subtyped into 6 $CD4^+$ and 7 $CD8^+$ clusters (Figures 2A, 2B, and S2A).[11] *PDCD1* (PD1) was uniformly expressed across cancer types by $CD8^+$ T_{EX}-cells, $CD4^+$ follicular-helper T cells (T_{FH}), and $CD4^+$ helper-1 T cells (T_{H1}), which correspond to subtypes consisting of clonally expanded, fully differentiated T cells.[12,13] These subtypes showed high expression of costimulatory molecules and various activation signatures, including exhaustion and stress response, and were metabolically active (Figures 2B and 2C). $CD8^+$ T_{EX}-cells were most frequent in MEL, while $CD4^+$ T_{FH}-cells were enriched in early BC and NSCLC (Figure S2B).

Since $CD8^+$ T_{EX}-cells and $CD4^+$ T_{REG}s are characterized by additional heterogeneity,[6] we further subclustered both subtypes. Particularly, 15,050 T_{EX}-cells subclustered into 6 subtypes, including $TCF7^+$, $GZMK^+$, terminal and proliferating T_{EX}-cells, as well as $CCL4^+$ and PKM^+ T_{EX}-cells not previously reported (Figures 2D and 2E). Although all $CD8^+$ T_{EX} subtypes expressed high levels of inhibitory checkpoints, terminal T_{EX}-cells expressed an additional subset of markers (*ENTPD1*, *CSF1*, and *LAYN*) previously associated with terminal exhaustion (Figures S2C and S2D).[6] $TCF7^+$ T_{EX}-cells expressed *TCF7*, a modulator of stem cell-like T cells and genes important for lymph node migration (*CCR7* and *SELL*). They exhibited a more naive phenotype (Figure 2E) and were enriched in MEL, as reported previously (Figure S2E).[14] PKM^+ T_{EX}-cells expressed high levels of glycolysis-related genes (*PKM*, *GAPDH*, and *ZBED2*), while also expressing checkpoint molecules (*LAG3* and *TIGIT*). $CCL4^+$ T_{EX}-cells showed the highest expression of various activation, T cell receptor (TCR) signaling and exhaustion signatures. Consequently, $CCL4^+$ T_{EX}-cells also expressed the highest levels of *PDCD1*.

Subclustering of 17,236 T_{REG}s revealed that they were either in a resting (*TNFRSF9*$^-$), intermediate (*ISG*$^+$) or activated state (*TNFRSF9*$^+$) (Figures 2F, 2G, and S2F).[6,15] $TNFRSF9^+$ T_{REG}s expressed various exhaustion markers (*CTLA4* and *TIGIT*), consistent with their increased activation status based on functional gene signatures (Figure S2G). $TNFRSF9^+$ T_{REG}s were enriched in HNSCC (Figure S2H). We also identified $S1PR1^+$ T_{REG}s characterized by naive signatures, as well as $CD8^+$ T_{REG}s defined by *CD3E*, *FOXP3*, and *CD8A* (but not *CD4*) expression (Figure S2I). They also expressed *ITGAE*, *NT5E*, and inhibitory molecules (*LAG3* and *FASLG*), while lacking *CD28* and *IL7R* expression, further distinguishing them from conventional $CD8^+$ T cells. Functionally, $CD8^+$ T_{REG}s secrete interleukin-10, which dampens immune responses. Consistent herewith, a significant enrichment of $CD8^+$ T_{REG}s was observed in GBM. $CD8^+$ T_{REG}s in our dataset also expressed cytotoxic markers such as GZMB, which according to prior findings is essential for their suppressive function.[16]

The B cell intra-tumoral landscape includes regulatory B cells

Among the 31,864 B cells, we identified 4 follicular B cell clusters (i.e., naive-mature B cells, germinal center [GC] B cells, and 2 memory B cell clusters) and 5 antibody-secreting plasma cells (i.e., plasmablasts and 2 IgG^+ and 2 IgA^+ plasma cell clusters; Figures 3A and 3B). Follicular B cells express surface markers associated with antigen presentation (major histocompatibility complex class II, *CD40*, *CD80*, and *CD86*; Figure S3A).[17] Compared to naive-mature B cells, memory B cells exhibit a more differentiated phenotype (*CD27*$^+$) and are known to populate the follicles of the spleen, lymph nodes and tumor-associated tertiary-lymphoid structures (TLSs).[18] Follicular B cells proliferate and undergo immunoglobulin class switching to give rise to plasmablasts ($IGHM^-$), which upon leaving the GC terminally differentiate into plasma cells. We identified 4 plasma cell clusters, which differed in terms of immunoglobulin heavy chain expression (*IGHG1* for IgG and *IGHA1* for IgA) and were either mature or immature based on antibody-secreting ability (high or low *PRDM1*) (Figure S3A). Finally, we also identified regulatory-memory B cells (B_{REG}s), which based on expression of inhibitory ligands and cytokines (*TGFB1* and *IL10*) are immunosuppressive and enriched in HNSCC (Figures 3B and S3A–S3C).[19] B_{REG}s indeed support tumor progression by inhibiting cytotoxic T cells and promoting T_{REG} proliferation.[20,21]

Early/inflammatory versus late/anti-inflammatory macrophages

Macrophages/monocytes (n = 45,821) consisted of 15 subtypes, including 1 neutrophil, 2 monocyte, and 12 macrophage (Mac) clusters (Figures 3C and 3D). Within the macrophages, monocyte-like macrophages (Mono-like Macs) and inflammatory macrophages (Inflam Macs) emerged as early macrophages closely related to monocytes. Mono-like Macs shared pro-inflammatory and monocyte-like profiles with classical monocytes (both expressing *CCR2*), while Inflam Macs stood out as the most inflamed, expressing high levels of *IL1B*, *CCL3/4*, and *CD274* (Figures 3C–3E, S3D, and S3E). The latter may represent the pro-inflammatory $PD\text{-}L1^+$ macrophages associated with

CellPress
OPEN ACCESS

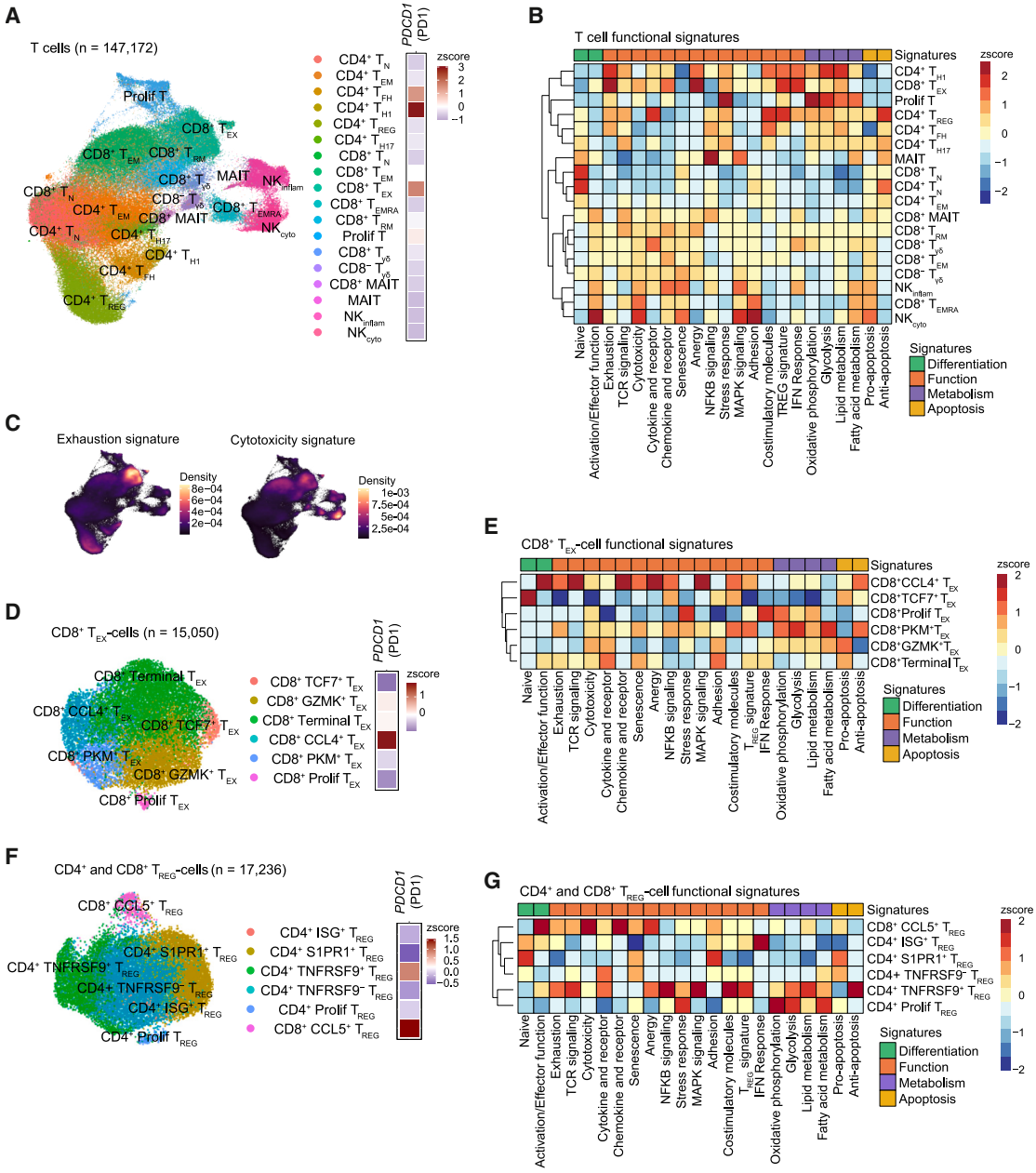


Figure 2. Heterogeneity within T cells, exhausted and regulatory T cells across cancer types

(A) Uniform manifold approximation and projection (UMAP; left) map of 147,172 T/NK cells subclustered into 18 subtypes as indicated by the color-coded legend and heatmap (right) of normalized *PD1* expression across T/NK cell subtypes.

(B) Heatmap showing the (scaled) expression of curated gene signatures (columns) across T/NK cell subtypes (rows).

(C) Density plots showing expression of exhaustion and cytotoxicity gene signatures.

(D) UMAP (left) of 15,050 $CD8^+$ T_{EX}-cells subclustered into 6 subtypes and heatmap (right) of normalized *PD1* expression across $CD8^+$ T_{EX}-cell subtypes.

(E) Heatmap showing the (scaled) expression of curated gene signatures (columns) across T_{EX}-cell subtypes (rows).

(F) UMAP (left) of 17,236 T_{REG}s subclustered into 6 subtypes and heatmap (right) of normalized *PD1* expression across T_{REG} subtypes.

(G) Heatmap showing the (scaled) expression of curated gene signatures (columns) across T_{REG} subtypes (rows).

*$p < 0.05$, **$p < 0.01$, and ***$p < 0.001$.

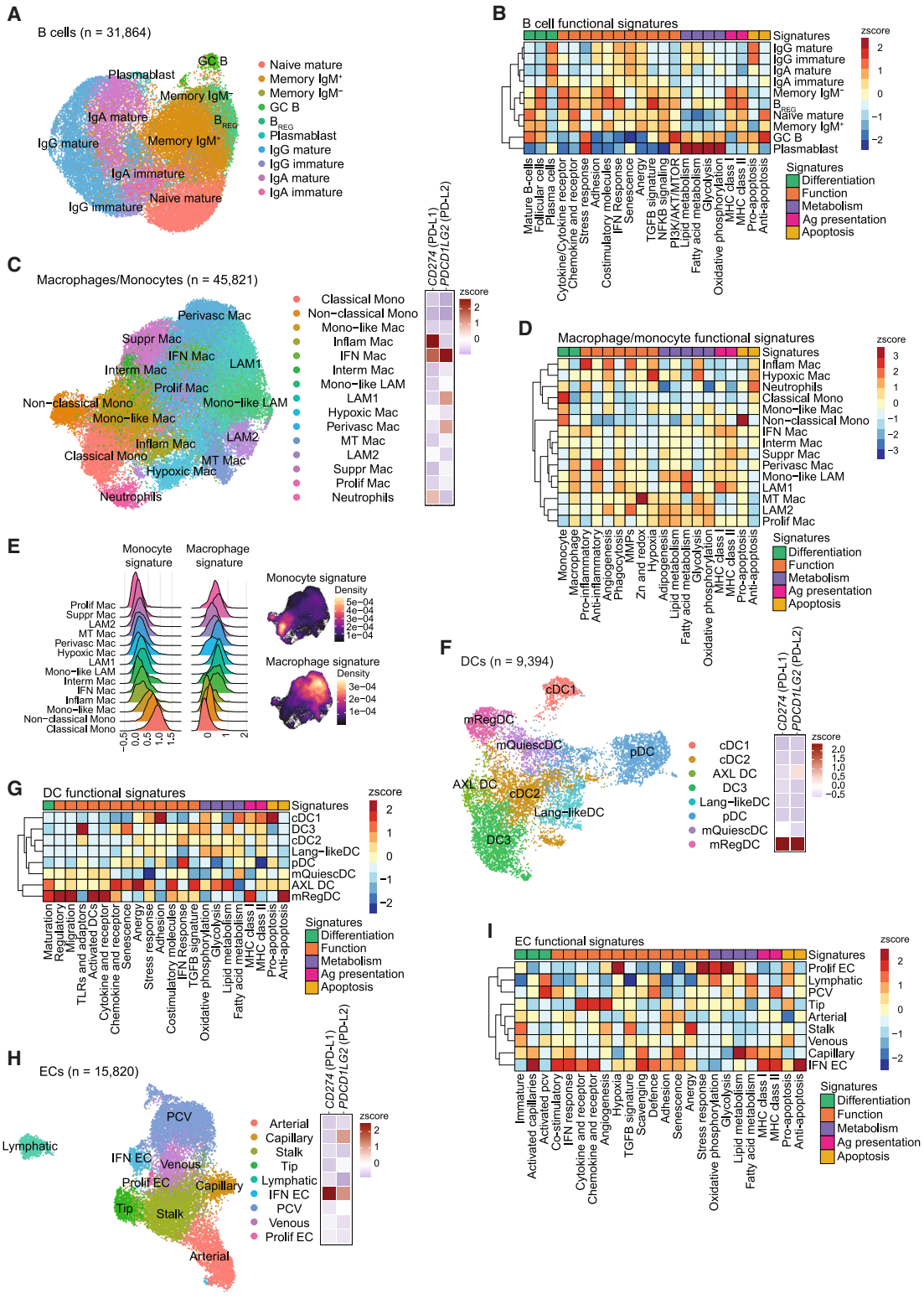


Figure 3. Heterogeneity within B cells, macrophages, DCs, and ECs across cancer types

(A) UMAP of 31,864 B cells subclustered into 10 subtypes as indicated by the color-coded legend.

(B) Heatmap showing (scaled) expression of curated gene signatures (columns) across B cell subtypes (rows).

(legend continued on next page)

immunostimulatory effects and favorable outcome in BC.[22] Interestingly, based on transcriptional signatures, Mono-like lipid-associated macrophages (Mono-like LAMs) represented a subtype intermediate to Inflam Macs and LAM1s. Similarly, intermediate macrophages (Interm Macs) were intermediate to Mono-like Macs and immunosuppressive Macs (Suppr Macs). While Mono-like and Inflam Macs expressed mainly pro-inflammatory markers, LAM1s, Mono-like LAMs and Suppr Macs expressed anti-inflammatory markers (*MSR1*, *CCL18*, and *AXL*), suggesting differentiation of pro-inflammatory early-macrophages into anti-inflammatory late-macrophages (Figures S3D and S3E). LAM1s were characterized by fatty acid metabolism, while LAM2s relative to LAM1s expressed higher levels of pro-inflammatory chemokines (*CXCL1*, *CXCL3*, and *CCL2*) (Figure S3F). Finally, Suppr Macs expressed genes involved in T cell inhibition, including *PDL4* and *TREM2*.

Consistent with macrophages being versatile in response to environmental cues, we could link other subtypes with distinct environments. For instance, interferon (IFN) Macs (*CXCL10* and *STAT1*) exhibited a strong type-1 IFN-mediated signature (Figure S3D), while neutrophils expressed pro-inflammatory (*CXCL8* and *IL1B*) and hypoxic genes (*VEGFA* and *ADM)*. These subtypes were most abundant in HNSCC (Figure S3G). In addition, both subtypes expressed *CD274*. On the opposite, macrophages in close contact with blood vessels, i.e., perivascular macrophages (Perivasc Macs) displayed a pronounced anti-inflammatory profile (Figure S3D). MT Macs expressed high levels of metallothioneins, which regulate zinc and redox factors to protect against oxidative stress.[23] Lastly, hypoxic Macs showed high expression of hypoxia, glycolysis and angiogenesis signatures.

The DC landscape includes different DCs subtypes and molecular states

DC clustering revealed 6 subtypes, including conventional type-1 and -2 (cDC1 and cDC2), two cDC2-related (DC3 and AXL DCs), plasmacytoid (pDC), and Langerhans-like DCs (Langh-like DC) (Figures 3F and 3G). In addition, we identified migratory DCs, which represent a cell state derived from cDC1s or cDC2s adopted when encountering cancer-associated antigens. MigDCs were identified based on *CCR7*, *CCL17*, and *CCL19* expression (Figure S3H),[24] and separated into ''quiescent'' (mQuiescDC) and ''mature DCs enriched for immunoregulatory molecules''. The latter additionally expressed genes associated with DC maturation, regulation, and migration (Figure S3G).[25] In line with their immune-modulatory role, mRegDCs expressed high levels of *CD274* (Figure 3F). Both MigDCs and AXL DCs were mainly present in HNSCC (Figure S3I). The latter are known to activate T cells and serve as potent allo-stimulatory cells, by expressing elevated levels of *CD86* and *HLA-DR*.[26] Our data confirm that AXL DCs express these markers, all of which are linked to T cell-mediated immune modulation (Figure S3H). In contrast, cDC1s play a critical role in fostering robust anti-tumor responses,[27] while cDC2s are involved in tumor immune evasion.[28] DC3s displayed a more pronounced pro-inflammatory profile when compared to AXL DCs, while maintaining transcriptional similarity with cDC2s.

A pan-cancer atlas of tumor endothelial cells

The subclustering of 15,820 ECs identified 9 subtypes (Figures 3H and 3I), including arterial and capillary ECs, stalk cells, tip cells, and lymphatic ECs. We also identified ECs characterized by antigen presentation and leukocyte trafficking (*IL6* and *CXCL8*), suggesting their involvement in immune cell recruitment (Figure S3J). Notably, these IFN ECs also expressed the highest levels of *CD274* within ECs (Figure 3H). Lymphatic ECs expressed high levels of *CCL21*, which is involved in the regulation of immune cell trafficking,[29] and were enriched in CRC (Figure S3K). Additionally, we distinguished venous from post-capillary venule (PCV) ECs (Figure S3L). The latter were particularly depleted in HCC and expressed typical PCV (*ACKR1* and *SELP*) and activation (*POSTN* and *CCL14*) markers, in line with their role as primary entry site of immune cells in organs.[30] Stalk cells expressed signatures of immature ECs and genes involved in angiogenesis (*FLT1* and *KDR*). This signature was also high in tip ECs. Finally, arterial ECs expressed *CXCL12*, a known marker of leukocyte trafficking.

Co-occurrences between TME cell subtypes reveal two distinct hubs

Next, we performed pairwise correlation analyses between cell subtype abundances to identify potential functional relationships within the TME. The analysis was performed on tumor samples only, since normal tissues and tLNs skew correlations due to an excess in T cells or B cells (Figures S4A and S4B). First, we examined how subtype abundances correlated with those of T cell subtypes. Within T cells, abundances of *PDCD1*-expressing T cells were all strongly correlated with each other (Figure 4A). For instance, $CD8^+$ T_{EX}-cells correlated with $CD4^+$ T_{FH}- and T_{H1}-cells. In contrast, *PDCD1*-expressing subtypes correlated negatively with naive T cell subtypes, including

(C) UMAP (left) of 45,821 macrophages/monocytes subclustered into 15 subtypes and heatmap (right) of normalized *PD-L1* and *PD-L2* expression across macrophage/monocyte subtypes.
(D) Heatmap showing (scaled) expression of curated gene signatures (columns) across macrophage/monocyte subtypes (rows).
(E) Ridge plots (left) and density plots (right) displaying expression of monocyte and macrophage signatures across macrophage/monocyte subtypes, ordered by increasing signature expression.
(F) UMAP (left) of 9,394 DCs subclustered into 8 subtypes as indicated by the color-coded legend and heatmap (right) of normalized *PD-L1* and *PD-L2* expression across DC subtypes.
(G) Heatmap showing (scaled) expression of curated gene signatures (columns) across DC subtypes (rows).
(H) UMAP (left) of 15,820 ECs subclustered into 9 subtypes as indicated by the color-coded legend and heatmap (right) of normalized *PD-L1* and *PD-L2* expression across EC subtypes.
(I) Heatmap showing (scaled) expression of curated gene signatures (columns) across EC subtypes (rows).
$^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

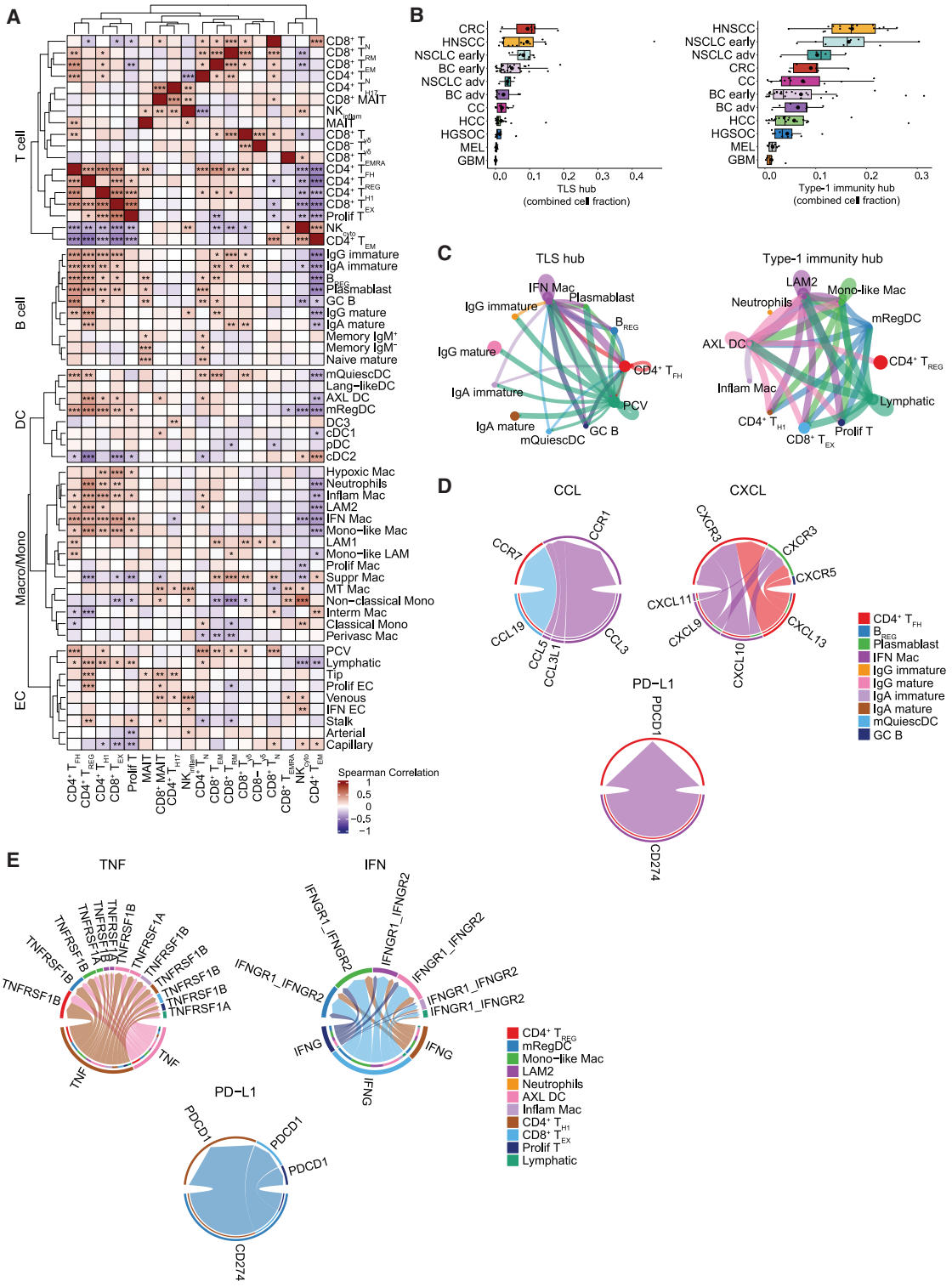


Figure 4. Co-occurring subtypes identify two distinct TME hubs

(A) Spearman correlations between relative abundances of T cell and B cell, macrophage/monocyte, and DC and EC subtypes per sample.

(B) Boxplots displaying the combined cell fraction of TLS hub (top) and type-1 immunity hub (bottom) across cancer types. All analyses displayed were performed on tumor samples with >500 cells (n = 125, see STAR Methods). Both hubs were frequent in CRC, as expected given that most samples were early-stage CRC, which typically responds to neoadjuvant ICB. Middle line: median; dot: mean; box edges: 25th and 75th percentiles.

(legend continued on next page)

$CD8^+$ T_N and $CD4^+$ T_{EM}-cells, as well as cytotoxic NKs. Vice versa, $CD8^+$ T_N-cells correlated positively with $CD4^+$ T_N and T_{EM}-cells. This suggests that different TMEs exist, one in which abundances of *PDCD1*-expressing clonally expanded, differentiated T cell subtypes prevail and another in which more naive T cell subtypes are frequent. Within B cells, plasma cells, plasmablasts, and B_{REG}s correlated positively with $CD4^+$ T_{FH}-cells (Figure 4A). Interestingly, all *PDCD1*-expressing T cells correlated positively with plasma cells, while the opposite was true for less-differentiated follicular B cells. In contrast, the latter were positively correlated with $CD4^+$ T_N-cells. This suggests that within a tumor, T cells and B cells follow differentiation trajectories that are closely intertwined with each other.

T-myeloid cell correlations showed early/inflammatory macrophages positively correlated with *PDCD1*-expressing T cells, but anti-correlated with $CD4^+$ T_{EM}-cells and $CD8^+$ T_N-cells, albeit not significantly for the latter (Figure 4A). Vice versa, Suppr Macs correlated negatively with *PDCD1*-expressing T cell subtypes, while they positively correlated with $CD8^+$ T_N- and T_{EM}-cells, and $CD4^+$ T_{EM}-cells. In DCs, mRegDCs and AXL DCs correlated positively with *PDCD1*-expressing T cell subtypes but negatively correlated with $CD4^+$ T_{EM}-cells and cytotoxic NKs. In contrast, cDC2s correlated negatively with T_{REG}s and *PDCD1*-expressing T cells, but positively with $CD4^+$ T_{EM}- and $CD8^+$ T_N-cells. Overall, these analyses reveal that some subtypes contribute to an immune-activated TME (characterized by *PDCD1*-expressing T cells), while others contribute to a more poorly immune-differentiated TME (characterized by more naive T cell subtypes).

To more comprehensively and unbiasedly dissect these TMEs, we explored co-occurrences between all subtypes simultaneously (Figure S4B). Hierarchical clustering identified 2 groups of positively correlating subtypes, which we refer to as "hubs." he first hub consisted of 10 subtypes, including differentiated B cells (i.e., the 4 plasma cell subtypes, plasmablasts, B_{REG}-, and GC B cells), IFN Macs, mQuiescDCs, and $CD4^+$ T_{FH}-cells. PCVs were also strongly correlated with the hub but based on hierarchical clustering did not formally belong to it. Since each of these subtypes are known to play a role in TLSs,[31–33] we refer to this hub as the "TLS hub." A second hub consisted of early/inflammatory macrophages (Mono-like Macs, Inflam Macs, neutrophils, and LAM2s), immune-regulatory cells (mRegDCs and $CD4^+$ T_{REG}s), as well as AXL DCs, lymphatic ECs, and importantly, *PDCD1*-expressing T cells ($CD4^+$ T_{H1}-, $CD8^+$ T_{EX}-, and proliferating T cells). Since both $CD4^+$ T_{H1}- and cytotoxic $CD8^+$ T_{EX}-cells are part of this hub, and also pro-inflammatory Macs and antigen-presenting DCs are involved, we refer to it as the "type-1 immunity hub."[34] Interestingly, both hubs were enriched in HNSCC and early NSCLC (Figure 4B).

Unraveling cell-cell interactions between subtypes within each hub

Next, we explored intercellular communication dynamics between the subtypes within each hub. Using CellChat,[35] we identified extensive interactions between all subtypes, i.e., 39 significant pathways in the tertiary lymphoid structures (TLS) and 72 in the type-1 immunity hub (Figures 4C and S4C). In the TLS hub, we detected prominent chemokine signaling, consistent with TLS development and maintenance relying on pro-inflammatory chemokines critical for immune cell migration.[31] For instance, we noticed *CCL19*:*CCR7* interactions between mQuiescDCs and $CD4^+$ T_{FH}-cells. *CXCL13*, essential for B cell homing to TLS, was produced by $CD4^+$ T_{FH}-cells and interacted with *CXCR5* on GC B cells (Figure 4D). Consistent with recent findings identifying other chemokines within the TLS,[36] we also observed *CCL3*, *CCL5*, *CXCL9*, *CXCL10*, and *CXCL11*, originating from IFN Macs and targeting *CXCR3* and *CXCR5* on $CD4^+$ T_{FH}-cells, GC B cells, and plasmablasts. Additionally, *CD274*:*PDCD1* interactions between IFN Macs and $CD4^+$ T_{FH}-cells were observed. In the type-1 immunity hub, we found that tumor necrosis factor alpha (TNF) and IFN signaling were important, whereby *TNF* and *INFG* ligands were expressed by $CD4^+$ T_{H1}-cells and other T cell subtypes, and their receptors (*TNFRSF1A/B* and *IFNGR1/2*) by macrophage and DC subtypes (Figure 4E). These findings highlight the complex immune interactions ongoing in both hubs.

The subtypes from each hub are spatially co-localized

Next, we explored whether correlations between cell subtypes within each hub are also reflected at the spatial level. We analyzed publicly available spatial transcriptomics datasets (Visium) from 60 samples selected from six studies (Table S4), covering various cancer types: BC (n = 6),[37] CRC (n = 12),[38,39] HCC (n = 7),[40] gastric cancer (GC; n = 9),[41] NSCLC (n = 20),[42] and MEL (n = 6).[43] After deconvoluting individual spots using gene signatures specific to each subtype in each hub (Table S3), we quantified their relative abundances in each of the Visium spots across the 60 samples. The correlation analysis of subtype scores within each spot revealed strong associations, indicating that they are spatially correlated (Figure 5A). This was confirmed across all six cancer types and in each cancer type separately (not shown). Overall, this suggests that the subtypes with correlated abundances in scRNA-seq data, are also spatially co-localized and form two spatially distinct hubs. We therefore generated signature scores for both hubs (Table S3), quantified them for each spot, and assessed their spatial localization across tumor sections. When examining the distribution of both hubs within each sample, we found that areas enriched for one hub did not necessarily overlap with those enriched for the other hub, though they were often in close proximity

(C) Circle plots showing the top 25% number of ligand(L)-receptor(R) interactions between pairwise subtypes belonging to TLS (top) and type-1 immunity (bottom) hubs. Edge width is proportional to the number of L-R pairs and edge colors are consistent with the signal sender. Circle sizes are proportional to the number of cells in each cell subtype.

(D and E) Chord diagrams illustrating cell-cell communication mediated by selected signaling pathways in TLS (D) and type-1 immunity (E) hubs. The inner, thinner bars indicate target cells receiving signals from the corresponding outer bars, with their colors representing the target identity. The size of the inner bars is proportional to the signal strength received by the targets. Individual L-R interactions are shown.

*p < 0.05, **p < 0.01, and ***p < 0.001.

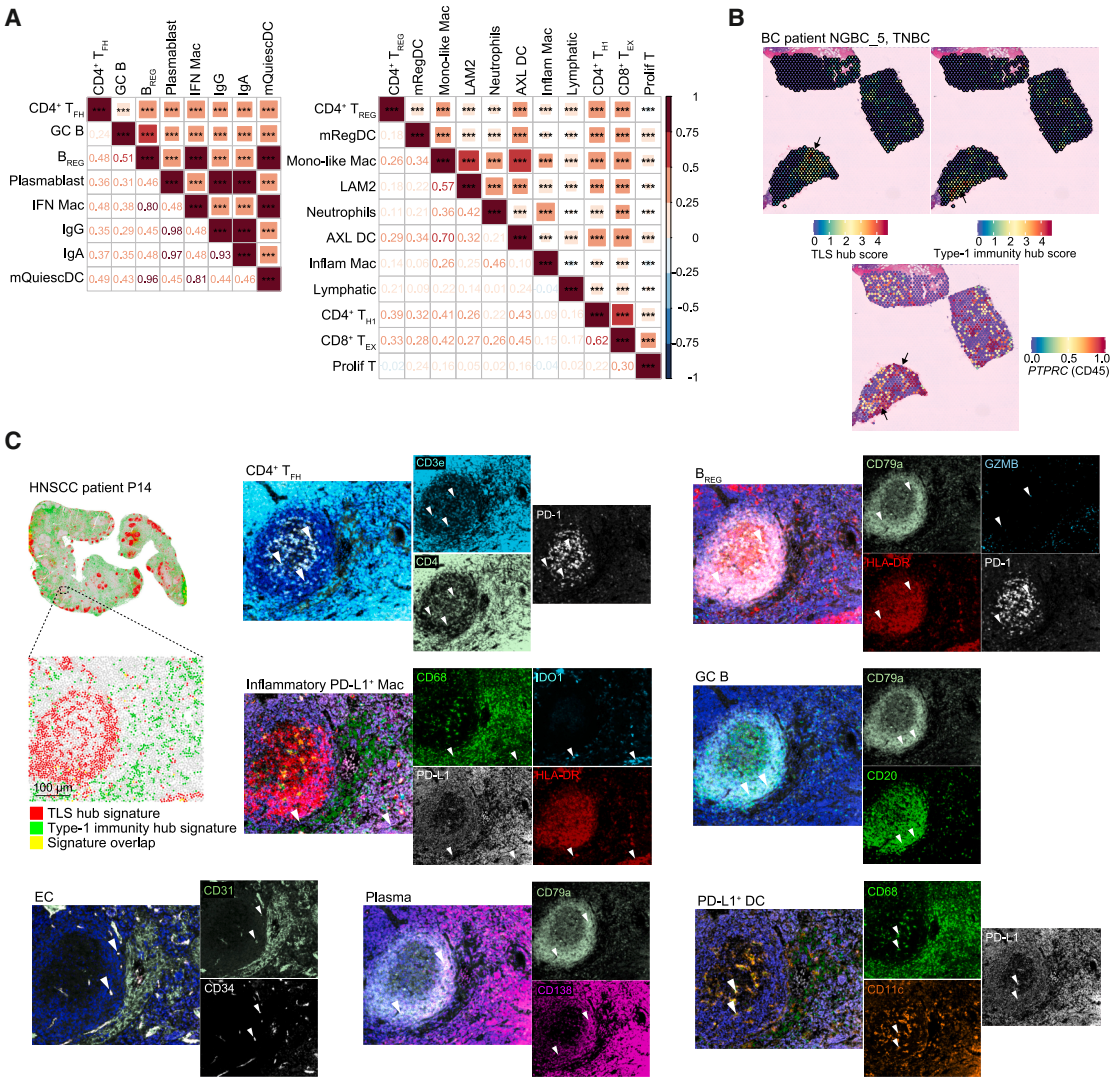


Figure 5. Spatial analysis of TME hubs across cancer types

(A) Pearson correlations of subtype signature scores in TLS (left) and type-1 immunity (right) hubs across 60 samples analyzed by Visium. Correlation values are shown.

(B) Scaled deconvolution values for TLS (top left) and type-1 immunity (top right) hubs overlaid onto tissue spots from the NGBC_5 triple negative BC (TNBC) sample.[37] *PTPRC* expression is shown as control (bottom). Black arrows highlight areas with high hub scores.

(C) Combined plot of TLS (red) and type-1 immunity (green) hub scores, with their overlap in yellow, in the HNSCC patient P14 sample analyzed with spatial proteomics (PhenoCycler, Akoya Biosciences). A close-up view highlights a region enriched for the TLS hub. Adjacent to this, a PhenoCycler-stained tumor section illustrates the co-localization of TLS hub-specific subtypes within the same region. For each subtype, the individual antibodies used for its identification, along with their merged signal, are displayed. White arrows indicate cells expressing the specified markers.

*$p < 0.05$, **$p < 0.01$, and ***$p < 0.001$.

(Figures 5B and S5A). We also checked immune cell localization within tumor sections by examining *PTPRC* (CD45). Importantly, while all hubs were found in *PTPRC*-enriched areas, not all *PTPRC*-enriched areas were high for the hubs, highlighting that they are specific and not just detecting areas enriched for immune cells.

Finally, these findings were validated using multiplex immunohistochemistry on HNSCC tumor sections (PhenoCycler).[44] Subtypes in each hub were identified based on the cumulative expression of epitopes targeted by 60 antibodies (see STAR Methods). While TLS hubs were discretely enriched, hubs for the type-1 immunity signature were more diffusely distributed, but frequently adjacent to the TLS hub (Figures 5C and S5B). When zooming in on an area with a high TLS signature score to examine the localization of specific subtypes, we identified CD4$^+$ T_{FH} cells (PD-1, CD3e, and CD4), plasma cells (CD79a and CD138), and GC B cells (CD79a and CD20), and a subset of B cells (CD79a) expressing PD-1, HLA-DR, and GZMB, which (based on scRNA-seq annotation) most likely correspond to B_{REG}s (Figure 5D). In the same region, we also observed

CellPress
OPEN ACCESS

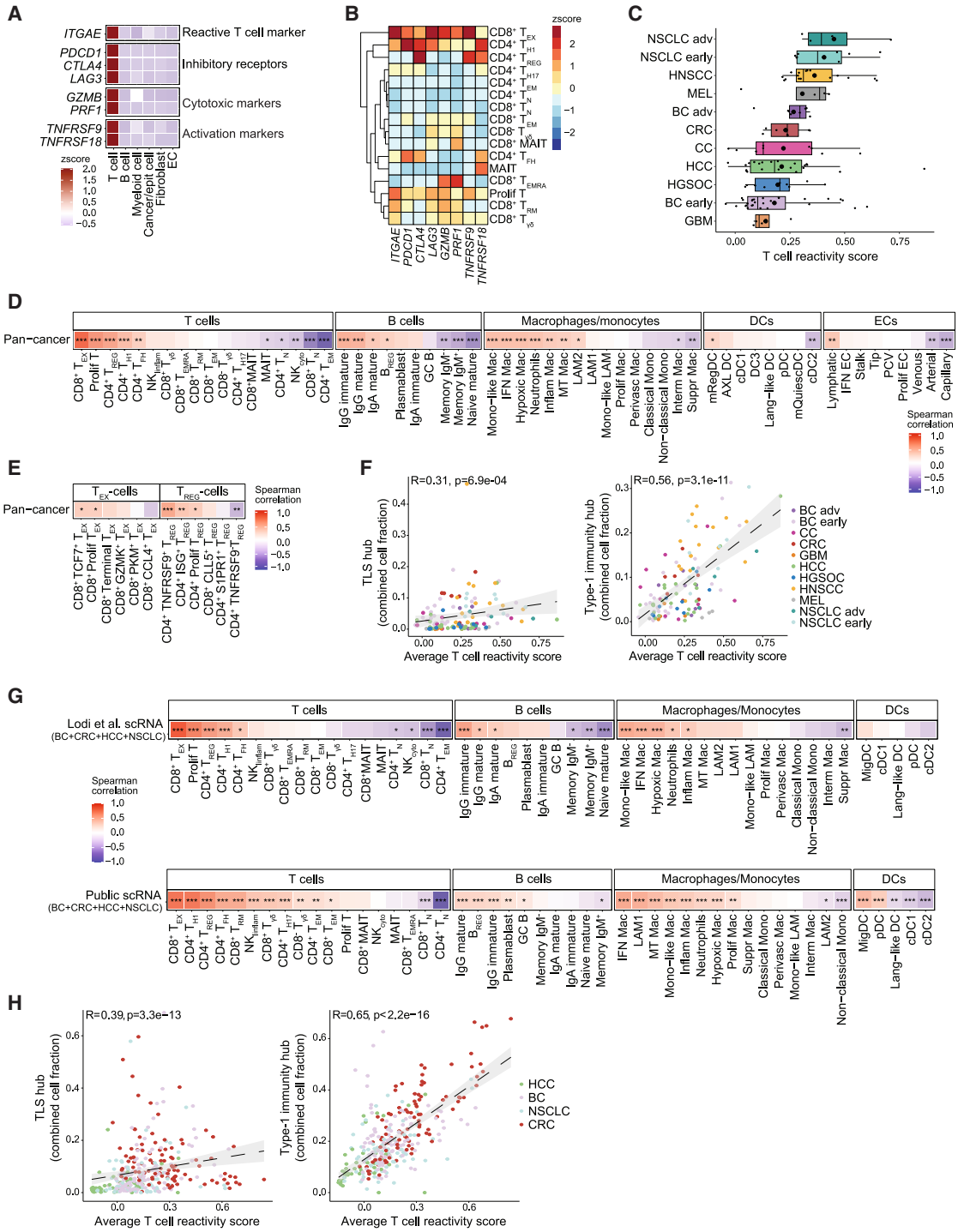


Figure 6. TME hubs associated with tumor-reactive T cells in pan-cancer datasets

(A) Heatmap showing normalized expression of genes included in the T cell reactivity signature across major cell types in tumor samples.

(B) Heatmap of gene expression of individual genes included in the T cell reactivity signature across T cell subtypes (rows) in tumor samples.

(C) Boxplots displaying expression of the T cell reactivity signature per sample across cancer types. Middle line: median; dot: mean; box edges: 25th and 75th percentiles.

(D and E) Spearman correlations between cell subtype abundances and T cell reactivity scores per sample at the pan-cancer level. The analysis was performed on tumor samples with enough cells (see STAR Methods) across T cell, B cell, macrophage/monocyte, DC and EC subtypes (D), as well as T_{EX} and T_{REG} subtypes (E). Subtypes are ranked from highest (left) to lowest (right) correlation scores in each cell type.

(legend continued on next page)

PD-L1$^+$ DCs (CD11c and CD68) representing MigDCs. Additionally, we identified cells positive for CD68, PD-L1, IDO1, and HLA-DR, which represent a pro-inflammatory macrophage population, as observed in our scRNA-seq data (Figure S3E). However, despite the 60 markers profiled, we could not distinguish whether this population represented IFN or Inflam Macs identified in the scRNA-seq data and therefore refer to them collectively as inflammatory macrophages. Likewise, due to limitations in antibody availability, we were unable to identify high endothelial venules (HEVs). However, we identified ECs (CD31 and CD34) surrounding the TLS hub, in close proximity to PD-1$^+$ T cells, suggesting that these cells may represent HEVs. In contrast, areas enriched for the type-1 immunity hub were strongly enriched for T cells (Figure S5B). Specifically, we identified CD4$^+$ T_{REG}s (CD3e, CD4, and FOXP3), CD8$^+$ T_{EX} (CD3e, CD8, PD-1, and IFNG), and CD4$^+$ T_{H1}-cells (CD3e, CD4, PD-1, and Tbx21). In the same region, we identified CD4$^+$ proliferating T cells (CD3e, CD4, and Ki67), neutrophils (MPO), along with inflammatory macrophages (CD68, PD-L1, IDO1, and HLA-DR) and DCs (PD-L1, CD68, and CD11c), closely resembling Inflam Macs and MigDCs, respectively (Figure S3E).

In summary, based on scRNA-seq, we identify various cell subtypes whose abundances are strongly correlated with each other, and which also spatially co-localize within 2 distinct TME hubs.

Subtypes correlating with T cell reactivity in the pan-cancer TME

Next, we explored which of these cell subtypes contribute to a tumor-reactive TME. We previously showed that T cells that become tumor-reactive upon ICB express inhibitory (*PDCD1*, *CTLA4*, and *LAG3*) and cytotoxic (*GZMB* and *PRF1*) markers already prior to the start of treatment.[11] We therefore derived a gene signature consisting of these genes, as well as tumor-reactive (*ITGAE*) and TCR activation markers (*TNFRSF9* and *TNFRSF18*) and refer to it as the T cell reactivity score[45] (Figure 6A). As expected, CD4$^+$ T_{H1}- and CD8$^+$ T_{EX}-cells, which are known to become tumor-reactive upon anti-PD1,[11] had the highest levels of this score (Figure 6B), while among cancer types NSCLC and HNSCC showed the highest score (Figure 6C). Next, we correlated the relative abundances of each cell subtype to the T cell reactivity scores within each tumor. We observed strong correlations with T cell reactivity for all *PDCD1*-expressing T cells and CD4$^+$ T_{REG}s, but negative correlations for naive T cell subtypes, including CD4$^+$ and CD8$^+$ T_N-cells, as well as CD4$^+$ T_{EM}-cells and cytotoxic NK cells (Figures 6D and S6A). Notably, within CD8$^+$ T_{EX}-cell subtypes, TCF7$^+$ and proliferating T_{EX}-cells showed the highest correlation, while within CD4$^+$ T_{REG}s, TNFRSF9$^+$, ISG$^+$, and proliferating T_{REG}s correlated with T cell reactivity, and TNFRSF9$^-$ T_{REG}s anti-correlated (Figures 6E and S6B). Differentiated B cell subtypes, i.e., IgG$^+$ or IgA$^+$ (im)mature plasma cells and plasmablasts showed positive correlations with T cell reactivity, while the naive-mature, memory IgM$^+$ and IgM$^-$ B cells correlated negatively. Interestingly, B_{REG}s also correlated positively. Among myeloid cells, early/inflammatory macrophages (e.g., Mono-like Macs and Inflam Macs), IFN Macs, and Hypoxic Macs correlated positively with T cell reactivity, while Suppr Macs and Interm Macs anti-correlated. Among DC subtypes, mRegDCs correlated positively, while cDC2s correlated negatively. Within ECs, lymphatic ECs exhibited a positive correlation, while arterial and capillary ECs anti-correlated. Since most of the subtypes positively correlating with T cell reactivity were also part of the 2 TME hubs, both hubs also correlated positively with T cell reactivity (Figure 6F).

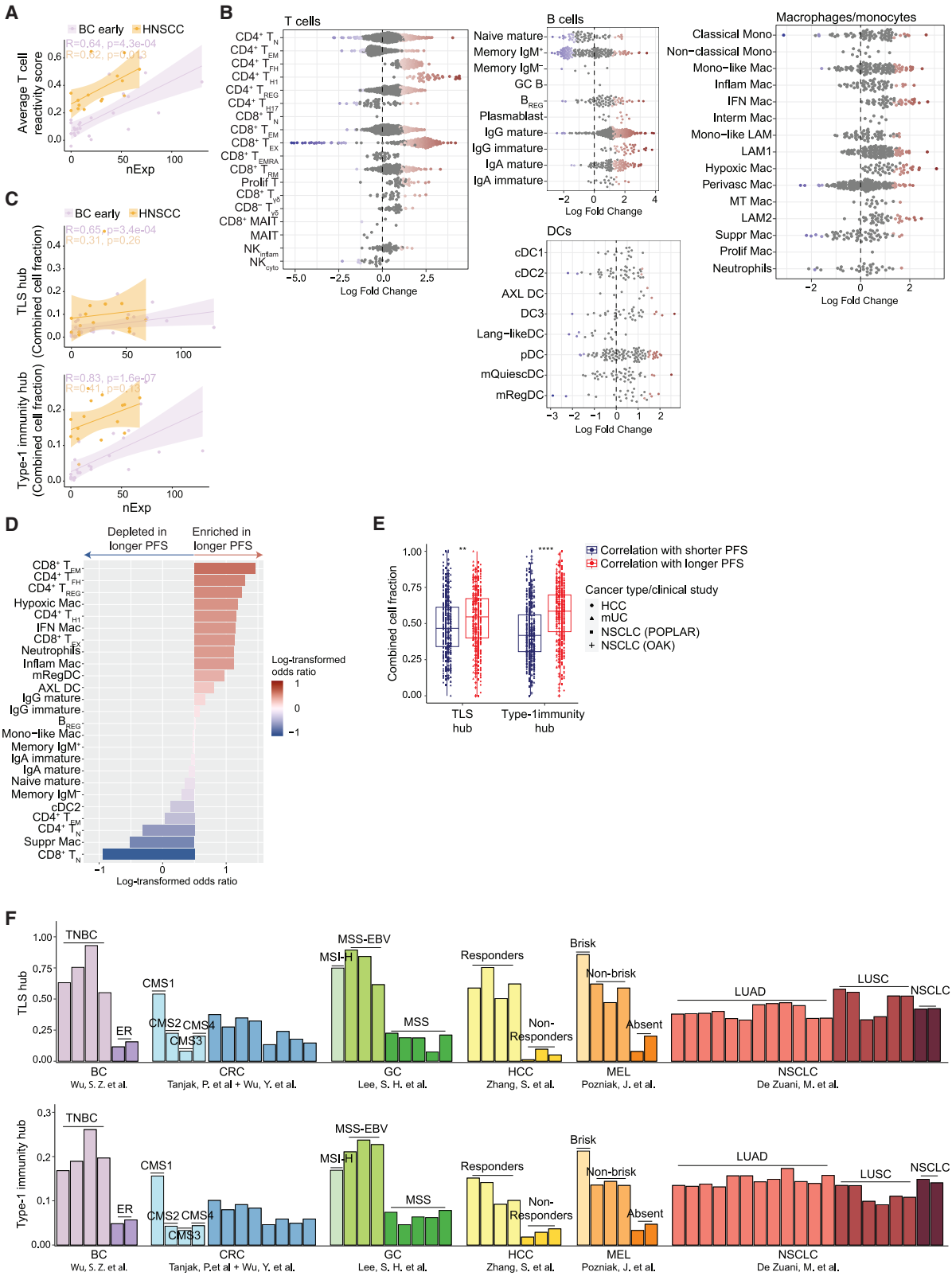
In summary, we found that PD1$^+$ T cells, immune-regulatory cells (CD4$^+$ T_{REG}s, B_{REG}s, and mRegDCs), plasma B cells, inflammatory/early-macrophages (Mono-like, IFN, and Inflam Macs) and Hypoxic Macs, LAM2s, neutrophils, and lymphatic ECs all positively correlated with T cell reactivity. The abundances of these subtypes were also correlated in our co-occurrence analyses, while most of them were also part of the 2 hubs.

Subtype abundance correlations in public datasets

To independently validate these observations, we searched for publicly available scRNA-seq datasets. To ensure that each cell type was proportionally represented, only studies using unbiasedly dissociated tumors were selected. Overall, 4, 3, 5, and 5 datasets from BC, CRC, HCC, and NSCLC respectively, were retrieved to create an atlas reflecting 1,320,145 cells profiled across 320 patient samples (BC, n = 104; CRC, n = 99; HCC, n = 47; NSCLC, n = 70)[4,5,11,37,46–58] (Table S4). For each cancer type, cell types were identified by marker gene annotation (Figure S6C), while cell subtypes were annotated by label transfer from our pan-cancer atlas. In the resulting publicly retrieved atlas, we could replicate all pairwise correlations between subtypes, but also the correlation between subtype abundances and T cell reactivity (Figure 6G). Finally, we observed that subtypes belonging to TLS or type-1 immunity hubs were significantly correlated with each other (Figure S6D), while both hubs also correlated positively with T cell reactivity (Figure 6H).

To assess the relevance of the hubs across even larger datasets, we computed hub-specific scores for each tumor sample from The Cancer Genome Atlas (TCGA) (n = 7,493 from 14 cancer types; Table S3). Consistent with our scRNA-seq findings, HNSCC showed the highest hub scores (Figure S6E), while the lowest scores were observed for (immunologically cold) glioma and prostate cancer. When assessing the relationship between the hubs and key clinical features, including mutation burden and immune cell infiltration (ImmCellInf; see STAR Methods) that represent key features of tumor-reactivity, both hub scores strongly correlated with ImmCellInf and mutation

(F) Spearman correlations between T cell reactivity and TLS (left) or type-1 immunity (right) hub combined cell fractions per sample.
(G) Spearman correlations between cell subtype abundances and T cell reactivity scores per sample in our dataset (top) and publicly available datasets (bottom; Table S4), across the indicated cancer types.
(H) Spearman correlations between T cell reactivity scores and TLS (left) or type-1 immunity (right) hub combined cell fractions per sample from publicly available datasets.
*p < 0.05, **p < 0.01, and ***p < 0.001.



(legend on next page)

burden (Figure S6F). Both hubs also correlated with signature scores from CD8$^+$ T_{EX}-cells (R = 0.84–0.89), but less with *PTPRC* expression (R = 0.71–0.64), highlighting their correlation with specific immune subtypes rather than all immune cells. Overall, these public scRNA-seq datasets reproduce key observations from our own atlas, while TCGA data reveal that TME hubs correlate with clinical features of an immune-activated TME.

TME subtypes and their hubs are associated with early response to ICB

Next, we explored whether subtypes correlating with T cell reactivity are also associated with response to ICB. We leveraged our scRNA-seq datasets generated in BC and HNSCC, in which patients after a pre-treatment tumor biopsy received pembrolizumab (anti-PD1) or durvalumab (anti-PD-L1), followed by an on-treatment biopsy 10 days later.[11,44] We defined T cell expansion as the increase in number of TCR clonotypes when comparing on- versus pre-treatment samples. First, we confirmed that high T cell reactivity in pre-treatment biopsies correlated with subsequent T cell expansion (Figures 7A and S7A). Consequently, differential abundance analysis of cell subtypes comparing expanding versus non-expanding tumors confirmed that subtypes associated with T cell reactivity were also enriched in tumors exhibiting expansion. Indeed, *PDCD1*-expressing T cells, such as CD4$^+$ T_{H1}, T_{FH}-, proliferating and CD8$^+$ T_{EX}-cells, as well as CD4$^+$ T_{REG}- and CD8$^+$ T_{EM}-cells were associated with expansion (Figures 7B and S7B). Cytotoxic NKs and CD4$^+$ T_{EM}-cells anti-correlated with expansion. Plasma cell subtypes, especially IgG mature plasma cells, were enriched in expanding tumors. In contrast, naive and memory IgM$^+$ B cells anti-correlated with expansion. Inflammatory/early-macrophages (Mono-like Macs, Inflam Macs, and IFN Macs), Hypoxic Macs, and LAM2s associated positively with T cell expansion, while classical monocytes and Suppr Macs correlated negatively (Figures 7B and S7B). mRegDCs, pDCs, cDC1s, and DC3s also correlated with expansion, but cDC2s appeared to anti-correlate. Finally, TLS and type-1 immunity hubs also correlated significantly with T cell expansion (Figures 7C and S7C). Collectively, these data indicate that the co-occurring subtypes that correlate with T cell reactivity and form the two hubs, were associated with T cell expansion early during ICB response.

TME subtypes and hubs associated with long-term clinical response to ICB

Next, we explored which subtypes correlated with long-term clinical outcome following ICB. For this, we analyzed bulk RNA-seq data from 4 clinical studies, in which tumor material was collected prior to receiving ICB (n = 937 patients: OAK in NSCLC, n = 344; POPLAR in NSCLC, n = 81; IMbrave150/GO30140 in HCC, n = 304; IMvigor210 in metastatic urothelial cancer or mUC, n = 208; Table S4).[60–63] For this we used Scissor,[64] which utilizes bulk RNA-seq data from these 937 patients to identify treatment response-associated cell subtypes within our single-cell atlas. Specifically, we calculated for each cell subtype the enrichment of cells associated with long versus short progression-free survival (PFS) across all studies. This revealed that within T cells, CD8$^+$ T_{EM} and T_{EX}, CD4$^+$ T_{FH} and T_{H1} subtypes were associated with long PFS, while CD4$^+$ and CD8$^+$ T_N- and CD4$^+$ T_{EM}-cells correlated with short PFS (Figures 7D and S7D). Among B cells, IgG mature and immature plasma cells associated with long PFS, while naive mature and memory IgM B cells associated with short PFS. Early/inflammatory macrophages, including IFN and Inflam Macs, as well as Hypoxic Macs and neutrophils, correlated with long PFS, while on the contrary, Suppr, MT, and Perivasc Macs correlated with short PFS. Within DCs, mRegDCs and AXL DCs correlated with long PFS, whereas cDC2s, but also cDC1s and Lang-like DCs correlated with short PFS. Interestingly, we also observed that both hubs correlated with prolonged PFS (Figure 7E). In the separate cancer types, the hubs were also associated with prolonged PFS (Figure S7E), except for HCC and CRC.

Finally, we analyzed our publicly available spatial datasets with respect to treatment response to ICB (Figure 7F). Triple negative BC (TNBC) exhibited higher scores for both hubs when compared to estrogen receptor (ER)-positive (ER$^+$) tumors, consistent with TNBC being the only BC subtype responsive to ICB.[65] In gastric cancer, MSI-H and Epstein-Barr virus (EBV)$^+$ samples showed the highest scores, aligning with their reported high immune cell infiltration.[66] HCC samples from patients who responded to ICB therapy had higher hub scores than non-responders. In MEL, samples were histopathologically classified based on immune infiltration levels as brisk (hot), non-brisk (immune-excluded), or cold, and our analysis confirmed that hot tumors had the highest scores. In NSCLC, all samples

Figure 7. TME hubs associated with clinical response to ICB across cancer types

(A) Spearman correlations between T cell reactivity scores calculated on T cells (excluding NK cells) and the number of expanded T cell clonotypes (nExp) in BC (n = 26) and HNSCC (n = 15). The analysis was performed on tumor samples with >500 cells and >20 T cells (see STAR Methods).

(B) Milo analysis in early BC and HNSCC showing differential subtype abundances between expanders (Es) and non-expanders (NEs) at a 5% false discovery rate (FDR). Red: enriched in Es, blue: enriched in NEs, gray: no significant enrichment.

(C) Spearman correlations between nExp and TLS (top) or type-1 immunity (bottom) hub combined cell fractions per sample in BC and HNSCC. The analysis was performed on samples with >500 cells and >20 T cells (see STAR Methods).

(D) Bar plot showing the enrichment of Scissor-positive cells associated with longer progression-free survival (PFS) across individual subtypes. Enrichment was quantified using log-transformed odds ratios of positive over total (positive plus negative) Scissor cells, calculated with the *Metafor* (v4.4-0)[59] R package. Average odds ratios across studies are shown.

(E) Boxplots showing the fraction of cells belonging to TLS or type-1 immunity hubs that are associated with long (red) or short (blue) PFS for each sample, as identified by Scissor. Scissor was performed against 4 clinical studies involving ICB with bulk RNA-seq data available, as indicated by the figure legend. Student's t test was performed per hub comparing fractions associated with long versus short PFS across the 4 clinical studies.

(F) Bar plots displaying average TLS (top) and type-1 immunity (bottom) hub scores across samples, color-coded by cancer type. The bar height represents the average hub scores across the spots from each sample.

GC, gastric cancer; mUC, metastatic urothelial cancer. $p < 0.05$, $^{**}p < 0.01$, and $^{***}p < 0.001$.

exhibited high hub scores, aligning with trials reporting that lung cancer responds to ICB.[60,61]

Data exploration across cancer types and shared subtypes

To encourage public data mining of our atlas, we created an interactive publicly available Shiny web[67] application: http://apps.lambrechtslab.sites.vib.be/PanCancer-Atlas. Besides being able to examine expression of individual genes or customized gene signatures in each cell (sub)type across cancer types or in each cancer type separately, it allows the ranking of individual tumors based on gene signature expression in any given cell type. Subsequently, it enables the correlation of cell subtype abundances with this ranking. Overall, this provides a unique means to explore TME characteristics associated with specific activated gene expression pathways at single-cell level.

DISCUSSION

We conducted a comprehensive analysis of 611,750 single cells from 9 cancer types across 160 patients. All samples were freshly collected and dissociated using a standardized protocol prior to scRNA-seq, minimizing the dissociation bias between cancer types. Furthermore, since cells were unbiasedly processed, we could assess heterogeneity within each cell type and directly compare it to that in other cell types. Overall, this allowed us to accurately assess how heterogeneity within each cell type contributes to an immune-active or -suppressive TME.

Our analysis identified up to 70 subtypes shared between cancer types. Among the $CD8^+$ T_{EX} subtypes, we identified 6 additional subtypes, 2 of which represented previously uncharacterized subtypes, i.e., $CCL4^+$ and PKM^+ T_{EX}-cells. Within $CD4^+$ T_{REG}s, we also recognized 6 additional subtypes. $CD8^+$ T_{EX}-cells are of particular interest because they exhibit high expression of PD1, while within $CD4^+$ T_{REG}s, some subtypes (i.e., $CD4^+$ $TNFRSF9^+$, $CD4^+$ ISG^+, and proliferating T_{REG}s) correlated with an immune-active, others ($TNFRSF9^-$ T_{REG}s) with an immune-suppressive TME. Interestingly, we also identified high PD1 expression in $CD4^+$ T_{H1}- and T_{FH}-cells, and PD-L1 in immune-regulatory subtypes, including mRegDCs and IFN ECs. Due to the pronounced expression of either PD1 or PD-L1, these subtypes might be involved in ICB response.

When exploring co-occurrences between subtypes, we identified two hubs for which subtype abundances were positively correlated with each other. The first hub included differentiated B cells such as B_{REG}s, 4 plasma cell subtypes, plasmablasts, $PD\text{-}L1^+$ IFN Macs, and $CD4^+$ T_{FH}-cells, all of which are involved in TLS. Indeed, the cellular composition of a TLS varies according to their maturation stage,[31] whereby a mature TLS consists of a segregated T cell zone containing $CXCL13^+$ $CD4^+$ T_{FH}-cells, mature DCs, and a B cell follicle with proliferating B cells, DCs, and macrophages.[33] Interestingly, PCVs also showed a clear positive correlation with most of these subtypes. This is intriguing because HEVs facilitate lymphocyte trafficking within lymph nodes, but are also present in intra-tumoral TLS.[32] Hence, although we could not distinguish HEVs from general PCVs in our scRNA-seq data, the cellular composition and its positive correlation with PCVs confirms this hub to represent TLS. The second hub of co-occurring subtypes consisted of early/inflammatory macrophages, neutrophils, $PD\text{-}L1^+$ and/or immune-regulatory cells, lymphatic ECs and $PD1^+$ T cells. The latter encompass the $CD8^+$ T_{EX} and $CD4^+$ T_{H1}-cells, which are known to clonally expand and exert anti-tumor responses during ICB.[11,44,68] Interestingly, spatial transcriptomic data of 60 tumors from 6 cancer types confirmed that the subtypes in each hub were also spatially co-localizing, illustrating that both hubs represent spatially defined TMEs.

We also explored the potential involvement of these subtypes during ICB response. First, we developed a T cell reactivity score that associated with response to ICB in pre-treatment tumors, and then analyzed which subtype abundances were associated with T cell reactivity. This revealed positive correlations with all subtypes within both hubs, while both hubs themselves also correlated with T cell reactivity across cancer types. Importantly, some subtypes also showed a negative correlation with T cell reactivity, including naive T cell and B cell subtypes, but also cytotoxic NKs, Suppr Macs, cDC2s, and arterial and capillary ECs. As such, our data provide unprecedented insights on how to interpret the functional contribution of each subtype to the TME. We also explored whether subtypes identified in both hubs correlate with early and long-term response to ICB. We considered T cell expansion after 1 dose of ICB in BC and HNSCC as a marker of early response to ICB, while across 937 patients participating in 4 clinical studies involving ICB, PFS was considered as long-term readout. Importantly, subtypes contributing to the 2 TME hubs, as well as the TME hubs themselves, were associated both with T cell expansion and improved PFS. In our spatial transcriptomic data, we observed that tumor sections enriched for both hubs were more frequent in tumors effectively responding to ICB, or in cancer types known to respond to ICB.

Limitations of the study

Finally, we developed a user-friendly Shiny app that enables comparisons between cancer types with respect to gene or gene signature expression, cell type abundances, their correlation with T cell reactivity, etc. It should be noted that for some cancer types, the number of samples included in the analysis is relatively small and comparative analyses within these cancer types should be interpreted with caution. The same applies for cell subtype associations with ICB response, which are better interpreted at pan-cancer level.

In conclusion, we present a comprehensive high-resolution single-cell atlas harmonized to facilitate comparisons in TME heterogeneity across cancer types. Our work serves as a valuable resource to explore TME heterogeneity in different biological contexts.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Diether Lambrechts (diether.lambrechts@kuleuven.be).

Materials availability

The study did not generate new unique reagents.

Data and code availability

- Read count data are publicly available at https://lambrechtslab.sites.vib.be/en/dataaccess. Raw sequencing reads are deposited under restricted access in the European Genome-Phenome Archive. Accession numbers for each dataset are listed in Table S1. Requests to access raw sequencing data will be reviewed by our data access committee on a per study basis. Any data shared will be released via a data transfer agreement including conditions guaranteeing protection of personal data according to European GDPR law. Accession numbers for publicly available datasets are in Table S4.
- All original code has been deposited in both GitHub (https://github.com/lambrechtslab/Pan-cancer_Lodietal) and Zenodo (https://doi.org/10.5281/zenodo.16848959), and is publicly available as of the date of publication.
- Any additional information required to reanalyze the data can be requested from the lead contact.

ACKNOWLEDGMENTS

This project was supported by VIB Grand Challenges. GBM and HCC by KU Leuven C-funding, GBM by a Marie Skłodowska-Curie grant (GLIOTRAIN) and Research Foundation Flanders (FWO), NSCLC and HCC by Stichting Tegen Kanker (STK), and NSCLC by M.D. Davidse Chair for immunology research in cancer. D.L. was supported by FWO (G065615N and G093821N), STK (C/2020/1529), the EU Innovation Program (RESCUER), KU Leuven C-funding (C14/22/125), and ERC (EXPAND IT). Computational resources were provided by the Flemish Supercomputer Center (VSC). The graphical abstract was created using BioRender.com and Adobe Illustrator 2024.

AUTHOR CONTRIBUTIONS

D.L. supervised experiments and wrote the manuscript with F.L. Data analysis was performed by F.L., S.V., and A.B. with significant contributions from B.B. Spatial analyses were performed by D.C., H.S., and T.V. Shiny app was created by L.F.M. and L.H. All authors have read and provided comments on the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS
  - Patient population
- METHOD DETAILS
  - Sample collection and processing
  - Single-cell RNA sequencing data acquisition and pre-processing
  - Single-cell gene expression analysis
  - Single-cell RNA sequencing clustering to determine major cell types
  - Comparison of dissociation biases in single-cell RNA sequencing and bulk RNA-seq across cancer types
  - Single-cell RNA sequencing clustering to determine cell subtypes
  - Curated functional gene signatures to characterize subtypes
  - Correlations between cell (sub)type abundances and T cell reactivity
  - Predicting cell-to-cell interactions in single-cell RNA sequencing data
  - Visium spatial transcriptomics data analysis
  - Akoya spatial proteomics data analysis
  - Public single-cell RNA sequencing datasets collection
  - T cell expansion analysis
  - Collection of public bulk RNA-seq datasets with immune checkpoint blockade response
  - Scissor analysis
  - The Cancer Genome Atlas analysis
- QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.xcrm.2025.102416.

Received: September 27, 2024
Revised: May 2, 2025
Accepted: September 19, 2025
Published: October 14, 2025

REFERENCES

1. Xu, L., Saunders, K., Huang, S.P., Knutsdottir, H., Martinez-Algarin, K., Terrazas, I., Chen, K., McArthur, H.M., Maués, J., Hodgdon, C., et al. (2024). A comprehensive single-cell breast tumor atlas defines epithelial and immune heterogeneity and interactions predicting anti-PD-1 therapy response. Cell Rep. Med. *5*, 101511.
2. Chung, W., Eum, H.H., Lee, H.O., Lee, K.M., Lee, H.B., Kim, K.T., Ryu, H.S., Kim, S., Lee, J.E., Park, Y.H., et al. (2017). Single-cell RNA-seq enables comprehensive tumour and immune cell profiling in primary breast cancer. Nat. Commun. *8*, 15081.
3. Leader, A.M., Grout, J.A., Maier, B.B., Nabet, B.Y., Park, M.D., Tabachnikova, A., Chang, C., Walker, L., Lansky, A., Le Berichel, J., et al. (2021). Single-cell analysis of human non-small cell lung cancer lesions refines tumor classification and patient stratification. Cancer Cell *39*, 1594–1609.e12.
4. Lu, Y., Yang, A., Quan, C., Pan, Y., Zhang, H., Li, Y., Gao, C., Lu, H., Wang, X., Cao, P., et al. (2022). A single-cell atlas of the multicellular ecosystem of primary and metastatic hepatocellular carcinoma. Nat. Commun. *13*, 4594.
5. Khaliq, A.M., Erdogan, C., Kurt, Z., Turgut, S.S., Grunvald, M.W., Rand, T., Khare, S., Borgia, J.A., Hayden, D.M., Pappas, S.G., et al. (2022). Refining colorectal cancer classification and clinical stratification through a single-cell atlas. Genome Biol. *23*, 113.
6. Zheng, L., Qin, S., Si, W., Wang, A., Xing, B., Gao, R., Ren, X., Wang, L., Wu, X., Zhang, J., et al. (2021). Pan-cancer single-cell landscape of tumor-infiltrating T cells. Science *374*, abe6474.
7. Chu, Y., Dai, E., Li, Y., Han, G., Pei, G., Ingram, D.R., Thakkar, K., Qin, J.J., Dang, M., Le, X., et al. (2023). Pan-cancer T cell atlas links a cellular stress response state to immunotherapy resistance. Nat. Med. *19*, 1–13.
8. Tang, F., Li, J., Qi, L., Liu, D., Bo, Y., Qin, S., Miao, Y., Yu, K., Hou, W., Li, J., et al. (2023). A pan-cancer single-cell panorama of human natural killer cells. Cell *186*, 4235–4251.e20. https://doi.org/10.1016/j.cell.2023.07.034.
9. Cheng, S., Li, Z., Gao, R., Xing, B., Gao, Y., Yang, Y., Qin, S., Zhang, L., Ouyang, H., Du, P., et al. (2021). A pan-cancer single-cell transcriptional atlas of tumor infiltrating myeloid cells. Cell *184*, 792–809.e23.
10. Luo, H., Xia, X., Huang, L.B., An, H., Cao, M., Kim, G.D., Chen, H.N., Zhang, W.H., Shu, Y., Kong, X., et al. (2022). Pan-cancer single-cell analysis reveals the heterogeneity and plasticity of cancer-associated fibroblasts in the tumor microenvironment. Nat. Commun. *131*, 6619.
11. Bassez, A., Vos, H., Van Dyck, L., Floris, G., Arijs, I., Desmedt, C., Boeckx, B., Vanden Bempt, M., Nevelsteen, I., Lambein, K., et al. (2021). A single-cell map of intratumoral changes during anti-PD1 treatment of patients with breast cancer. Nat. Med. *275*, 820–832.
12. Petrelli, A., Mijnheer, G., Hoytema van Konijnenburg, D.P., van der Wal, M.M., Giovannone, B., Mocholi, E., Vazirpanah, N., Broen, J.C., Hijnen, D.,

Oldenburg, B., et al. (2018). PD-1+CD8+ T cells are clonally expanding effectors in human chronic inflammation. J. Clin. Investig. *128*, 4669–4681.

13. Wu, T.D., Madireddi, S., de Almeida, P.E., Banchereau, R., Chen, Y.J.J., Chitre, A.S., Chiang, E.Y., Iftikhar, H., O'Gorman, W.E., Au-Yeung, A., et al. (2020). Peripheral T cell expansion predicts tumour infiltration and clinical response. Nature *579*, 274–278.
14. Sade-Feldman, M., Yizhak, K., Bjorgaard, S.L., Ray, J.P., de Boer, C.G., Jenkins, R.W., Lieb, D.J., Chen, J.H., Frederick, D.T., Barzily-Rokni, M., et al. (2018). Defining T Cell States Associated with Response to Checkpoint Immunotherapy in Melanoma. Cell *175*, 998–1013.e20.
15. Cho, J.W., Son, J., Ha, S.J.,, and Lee, I. (2021). Systems biology analysis identifies TNFRSF9 as a functional marker of tumor-infiltrating regulatory T-cell enabling clinical outcome prediction in lung cancer. Comput. Struct. Biotechnol. J. *19*, 860–868.
16. Chen, X., Ghanizada, M., Mallajosyula, V., Sola, E., Capasso, R., Kathuria, K.R., and Davis, M.M. (2025). Differential roles of human CD4+ and CD8+ regulatory T cells in controlling self-reactive immune responses. Nat. Immunol. *26*, 230–239.
17. Bruno, T.C., Ebner, P.J., Moore, B.L., Squalls, O.G., Waugh, K.A., Eruslanov, E.B., Singhal, S., Mitchell, J.D., Franklin, W.A., Merrick, D.T., et al. (2017). Antigen-presenting intratumoral B cells affect CD4+ TIL phenotypes in non–small cell lung cancer patients. Cancer Immunol. Res. *5*, 898–907.
18. Zhang, Q., and Wu, S. (2022). Tertiary lymphoid structures are critical for cancer prognosis and therapeutic response. Front. Immunol. *13*, 1063711.
19. Balkwill, F., Montfort, A.,, and Capasso, M. (2013). B regulatory cells in cancer. Trends Immunol. *34*, 169–173.
20. Schwartz, M., Zhang, Y.,, and Rosenblatt, J.D. (2016). B cell regulation of the anti-tumor response and role in carcinogenesis. J. Immunother. Cancer *4*, 40. https://doi.org/10.1186/s40425-016-0145-x.
21. Olkhanud, P.B., Damdinsuren, B., Bodogai, M., Gress, R.E., Sen, R., Wejksza, K., Malchinkhuu, E., Wersto, R.P., and Biragyn, A. (2011). Tumor-evoked regulatory B cells promote breast cancer metastasis by converting resting CD4+ T cells to T-regulatory cells. Cancer Res. *71*, 3505–3515.
22. Wang, L., Guo, W., Guo, Z., Yu, J., Tan, J., Simons, D.L., Hu, K., Liu, X., Zhou, Q., Zheng, Y., et al. (2024). PD-L1-expressing tumor-associated macrophages are immunostimulatory and associate with good clinical outcome in human breast cancer. Cell Rep. Med. *5*, 101420.
23. Serra, M., Columbano, A., Ammarah, U., Mazzone, M.,, and Menga, A. (2020). Understanding Metal Dynamics Between Cancer Cells and Macrophages: Competition or Synergism? Front. Oncol. *10*, 646.
24. Ginhoux, F., Guilliams, M.,, and Merad, M. (2022). Expanding dendritic cell nomenclature in the single-cell era. Nat. Rev. Immunol. *22*, 67–68.
25. Maier, B., Leader, A.M., Chen, S.T., Tung, N., Chang, C., LeBerichel, J., Chudnovskiy, A., Maskey, S., Walker, L., Finnigan, J.P., et al. (2020). A conserved dendritic-cell regulatory program limits antitumour immunity. Nature *580*, 257–262.
26. Rhodes, J.W., Tong, O., Harman, A.N.,, and Turville, S.G. (2019). Human Dendritic Cell Subsets, Ontogeny, and Impact on HIV Infection. Front. Immunol. *10*, 1088.
27. Shapir Itai, Y., Barboy, O., Salomon, R., Bercovich, A., Xie, K., Winter, E., Shami, T., Porat, Z., Erez, N., Tanay, A., et al. (2024). Bispecific dendritic-T cell engager potentiates anti-tumor immunity. Cell *187*, 375–389.e18.
28. Saito, Y., Komori, S., Kotani, T., Murata, Y.,, and Matozaki, T. (2022). The Role of Type-2 Conventional Dendritic Cells in the Regulation of Tumor Immunity. Cancers *14*, 1976.
29. Tewalt, E.F., Cohen, J.N., Rouhani, S.J.,, and Engelhard, V.H. (2012). Lymphatic endothelial cells - key players in regulation of tolerance and immunity. Front. Immunol. *3*, 305.
30. Goveia, J., Rohlenova, K., Taverna, F., Treps, L., Conradi, L.C., Pircher, A., Geldhof, V., de Rooij, L.P.M.H., Kalucka, J., Sokol, L., et al. (2020). An Integrated Gene Expression Landscape Profiling Approach to Identify Lung Tumor Endothelial Cell Heterogeneity and Angiogenic Candidates. Cancer Cell *37*, 21–36.e13.
31. Schumacher, T.N., and Thommen, D.S. (2022). Tertiary lymphoid structures in cancer. Science *375*, eabf9419.
32. Vella, G., Hua, Y.,, and Bergers, G. (2023). High endothelial venules in cancer: Regulation, function, and therapeutic implication. Cancer Cell *41*, 527–545.
33. Garaud, S., Dieu-Nosjean, M.-C.,, and Willard-Gallo, K. (2022). T follicular helper and B cell crosstalk in tertiary lymphoid structures and cancer immunotherapy. Nat. Commun. *13*, 2259.
34. Annunziato, F., Romagnani, C.,, and Romagnani, S. (2015). The 3 major types of innate and adaptive cell-mediated effector immunity. J. Allergy Clin. Immunol. *135*, 626–635.
35. Jin, S., Guerrero-Juarez, C.F., Zhang, L., Chang, I., Ramos, R., Kuan, C.H., Myung, P., Plikus, M.V., and Nie, Q. (2021). Inference and analysis of cell-cell communication using CellChat. Nat. Commun. *12*, 1088.
36. Fan, X., Feng, D., Wei, D., Li, A., Wei, F., Deng, S., Shen, M., Qin, C., Yu, Y., and Liang, L. (2025). Characterizing tertiary lymphoid structures associated single-cell atlas in breast cancer patients. Cancer Cell Int. *25*, 12.
37. Wu, S.Z., Al-Eryani, G., Roden, D.L., Junankar, S., Harvey, K., Andersson, A., Thennavan, A., Wang, C., Torpy, J.R., Bartonicek, N., et al. (2021). A single-cell and spatially resolved atlas of human breast cancers. Nat. Genet. *53*, 1334–1347.
38. Wu, Y., Yang, S., Ma, J., Chen, Z., Song, G., Rao, D., Cheng, Y., Huang, S., Liu, Y., Jiang, S., et al. (2022). Spatiotemporal Immune Landscape of Colorectal Cancer Liver Metastasis at Single-Cell Level. Cancer Discov. *12*, 134–153.
39. Tanjak, P., Chaiboonchoe, A., Suwatthanarak, T., Thanormjit, K., Acharayothin, O., Chanthercrob, J., Parakonthun, T., Methasate, A., Fischer, J.M., Wong, M.H., and Chinswangwatanakul, V. (2024). Tumor-immune hybrid cells evade the immune response and potentiate colorectal cancer metastasis through CTLA4. Clin. Exp. Med. *25*, 2.
40. Zhang, S., Yuan, L., Danilova, L., Mo, G., Zhu, Q., Deshpande, A., Bell, A.T. F., Elisseeff, J., Popel, A.S., Anders, R.A., et al. (2023). Spatial transcriptomics analysis of neoadjuvant cabozantinib and nivolumab in advanced hepatocellular carcinoma identifies independent mechanisms of resistance and recurrence. Genome Med. *15*, 72.
41. Lee, S.H., Lee, D., Choi, J., Oh, H.J., Ham, I.H., Ryu, D., Lee, S.Y., Han, D. J., Kim, S., Moon, Y., et al. (2025). Spatial dissection of tumour microenvironments in gastric cancers reveals the immunosuppressive crosstalk between CCL2+ fibroblasts and STAT3-activated macrophages. Gut *74*, 714–727. https://doi.org/10.1136/gutjnl-2024-332901.
42. De Zuani, M., Xue, H., Park, J.S., Dentro, S.C., Seferbekova, Z., Tessier, J., Curras-Alonso, S., Hadjipanayis, A., Athanasiadis, E.I., Gerstung, M., et al. (2024). Single-cell and spatial transcriptomics analysis of non-small cell lung cancer. Nat. Commun. *15*, 4388.
43. Pozniak, J., Pedri, D., Landeloos, E., Van Herck, Y., Antoranz, A., Vanwynsberghe, L., Nowosad, A., Roda, N., Makhzami, S., Bervoets, G., et al. (2024). A TCF4-dependent gene regulatory network confers resistance to immunotherapy in melanoma. Cell *187*, 166–183.e25.
44. Franken, A., Bila, M., Mechels, A., Kint, S., Van Dessel, J., Pomella, V., Vanuytven, S., Philips, G., Bricard, O., Xiong, J., et al. (2024). CD4+ T cell activation distinguishes response to anti-PD-L1+anti-CTLA4 therapy from anti-PD-L1 monotherapy. Immunity *57*, 541–558.e7.
45. van der Leun, A.M., Thommen, D.S.,, and Schumacher, T.N. (2020). CD8+ T cell states in human cancer: insights from single-cell analysis. Nat. Rev. Cancer *20*, 218–232.
46. Metoikidou, C., Karnaukhov, V., Boeckx, B., Timperi, E., Bonté, P.E., Wang, L., Espenel, M., Albaud, B., Loirat, D., Wang, X., et al. (2025). Continuous replenishment of the dysfunctional CD8 T cell axis is associated with response to chemoimmunotherapy in advanced breast cancer. Cell Rep. Med. *18*, 101973.

47. Pal, B., Chen, Y., Vaillant, F., Capaldo, B.D., Joyce, R., Song, X., Bryant, V. L., Penington, J.S., Di Stefano, L., Tubau Ribera, N., et al. (2021). A single-cell RNA expression atlas of normal, preneoplastic and tumorigenic states in the human breast. EMBO J. *40*, e107333.

48. Pelka, K., Hofree, M., Chen, J.H., Sarkizova, S., Pirl, J.D., Jorgji, V., Bejnood, A., Dionne, D., Ge, W.H., Xu, K.H., et al. (2021). Spatially organized multicellular immune hubs in human colorectal cancer. Cell *184*, 4734–4752.e20.

49. Lee, H.-O., Hong, Y., Etlioglu, H.E., Cho, Y.B., Pomella, V., Van den Bosch, B., Vanhecke, J., Verbandt, S., Hong, H., Min, J.W., et al. (2020). Lineage-dependent gene expression programs influence the immune landscape of colorectal cancer. Nat. Genet. *52*, 594–603.

50. Losic, B., Craig, A.J., Villacorta-Martin, C., Martins-Filho, S.N., Akers, N., Chen, X., Ahsen, M.E., von Felden, J., Labgaa, I., D'Avola, D., et al. (2020). Intratumoral heterogeneity and clonal evolution in liver cancer. Nat. Commun. *11*, 291.

51. Ma, L., Wang, L., Khatib, S.A., Chang, C.W., Heinrich, S., Dominguez, D. A., Forgues, M., Candia, J., Hernandez, M.O., Kelly, M., et al. (2021). Single-cell atlas of tumor cell evolution in response to therapy in hepatocellular carcinoma and intrahepatic cholangiocarcinoma. J. Hepatol. *75*, 1397–1408.

52. Ho, D.W.-H., Tsui, Y.M., Chan, L.K., Sze, K.M.F., Zhang, X., Cheu, J.W.S., Chiu, Y.T., Lee, J.M.F., Chan, A.C.Y., Cheung, E.T.Y., et al. (2021). Single-cell RNA sequencing shows the immunosuppressive landscape and tumor heterogeneity of HBV-associated hepatocellular carcinoma. Nat. Commun. *12*, 3684.

53. Ma, L., Heinrich, S., Wang, L., Keggenhoff, F.L., Khatib, S., Forgues, M., Kelly, M., Hewitt, S.M., Saif, A., Hernandez, J.M., et al. (2022). Multiregional single-cell dissection of tumor and immune cells reveals stable lock-and-key features in liver cancer. Nat. Commun. *13*, 7533.

54. Bischoff, P., Trinks, A., Obermayer, B., Pett, J.P., Wiederspahn, J., Uhlitz, F., Liang, X., Lehmann, A., Jurmeister, P., Elsner, A., et al. (2021). Single-cell RNA sequencing reveals distinct tumor microenvironmental patterns in lung adenocarcinoma. Oncogene *40*, 6748–6758.

55. Kim, N., Kim, H.K., Lee, K., Hong, Y., Cho, J.H., Choi, J.W., Lee, J.I., Suh, Y.L., Ku, B.M., Eum, H.H., et al. (2020). Single-cell RNA sequencing demonstrates the molecular and cellular reprogramming of metastatic lung adenocarcinoma. Nat. Commun. *11*, 2285.

56. He, D., Wang, D., Lu, P., Yang, N., Xue, Z., Zhu, X., Zhang, P., and Fan, G. (2021). Single-cell RNA sequencing reveals heterogeneous tumor and immune cell populations in early-stage lung adenocarcinomas harboring EGFR mutations. Oncogene *40*, 355–368.

57. Laughney, A.M., Hu, J., Campbell, N.R., Bakhoum, S.F., Setty, M., Lavallée, V.P., Xie, Y., Masilionis, I., Carr, A.J., Kottapalli, S., et al. (2020). Regenerative lineages and immune-mediated pruning in lung cancer metastasis. Nat. Med. *26*, 259–269.

58. Chen, J., Tan, Y., Sun, F., Hou, L., Zhang, C., Ge, T., Yu, H., Wu, C., Zhu, Y., Duan, L., et al. (2020). Single-cell transcriptome and antigen-immunoglobin analysis reveals the diversity of B cells in non-small cell lung cancer. Genome Biol. *21*, 152.

59. Viechtbauer, W. (2010). Conducting Meta-Analyses in R with the metafor Package. J. Stat. Softw. *36*, 1–48.

60. Fehrenbacher, L., Spira, A., Ballinger, M., Kowanetz, M., Vansteenkiste, J., Mazieres, J., Park, K., Smith, D., Artal-Cortes, A., Lewanski, C., et al. (2016). Atezolizumab versus docetaxel for patients with previously treated non-small-cell lung cancer (POPLAR): a multicentre, open-label, phase 2 randomised controlled trial. Lancet *387*, 1837–1846.

61. Rittmeyer, A., Barlesi, F., Waterkamp, D., Park, K., Ciardiello, F., von Pawel, J., Gadgeel, S.M., Hida, T., Kowalski, D.M., Dols, M.C., et al. (2017). Atezolizumab versus docetaxel in patients with previously treated non-small-cell lung cancer (OAK): a phase 3, open-label, multicentre randomised controlled trial. Lancet *389*, 255–265.

62. Vander Velde, N., Guerin, A., Ionescu-Ittu, R., Shi, S., Wu, E.Q., Lin, S.W., Hsu, L.I., Saum, K.U., de Ducla, S., and Wang, J. (2019). Comparative effectiveness of non-cisplatin first-line therapies for metastatic urothelial carcinoma: Phase 2 IMvigor210 study versus US patients treated in the veterans health administration. Eur. Urol. Oncol. *2*, 12–20. https://doi.org/10.1016/j.euo.2018.07.003.

63. Lee, M.S., Ryoo, B.Y., Hsu, C.H., Numata, K., Stein, S., Verret, W., Hack, S.P., Spahn, J., Liu, B., Abdullah, H., et al. (2020). Atezolizumab with or without bevacizumab in unresectable hepatocellular carcinoma (GO30140): an open-label, multicentre, phase 1b study. Lancet Oncol. *21*, 808–820.

64. Sun, D., Guan, X., Moran, A.E., Wu, L.Y., Qian, D.Z., Schedin, P., Dai, M.S., Danilov, A.V., Alumkal, J.J., Adey, A.C., et al. (2022). Identifying phenotype-associated subpopulations by integrating bulk and single-cell sequencing data. Nat. Biotechnol. *40*, 527–538.

65. Rayson, V.C., Harris, M.A., Savas, P., Hun, M.L., Virassamy, B., Salgado, R., and Loi, S. (2024). The anti-cancer immune response in breast cancer: current and emerging biomarkers and treatments. Trends Cancer *10*, 490–506.

66. Kang, B.W., Seo, A.N., Yoon, S., Bae, H.I., Jeon, S.W., Kwon, O.K., Chung, H.Y., Yu, W., Kang, H., and Kim, J.G. (2016). Prognostic value of tumor-infiltrating lymphocytes in Epstein–Barr virus-associated gastric cancer. Ann. Oncol. *27*, 494–501.

67. Ouyang, J.F., Kamaraj, U.S., Cao, E.Y.,, and Rackham, O.J.L. (2021). ShinyCell: simple and sharable visualization of single-cell gene expression data. Bioinformatics *37*, 3374–3376.

68. Yu, R., Zhu, B.,, and Chen, D. (2022). Type I interferon-mediated tumor immunity and its role in immunotherapy. Cell. Mol. Life Sci. *79*, 191.

69. Hao, Y., Hao, S., Andersen-Nissen, E., Mauck, W.M., 3rd, Zheng, S., Butler, A., Lee, M.J., Wilk, A.J., Darby, C., Zager, M., et al. (2021). Integrated analysis of multimodal single-cell data. Cell *184*, 3573–3587.e29.

70. McGinnis, C.S., Murrow, L.M.,, and Gartner, Z.J. (2019). DoubletFinder: Doublet Detection in Single-Cell RNA Sequencing Data Using Artificial Nearest Neighbors. Cell Syst. *8*, 329–337.e4.

71. Korsunsky, I., Millard, N., Fan, J., Slowikowski, K., Zhang, F., Wei, K., Baglaenko, Y., Brenner, M., Loh, P.R., and Raychaudhuri, S. (2019). Fast, sensitive and accurate integration of single-cell data with Harmony. Nat. Methods *1612*, 1289–1296.

72. Cappuyns, S., Philips, G., Vandecaveye, V., Boeckx, B., Schepers, R., Van Brussel, T., Arijs, I., Mechels, A., Bassez, A., Lodi, F., et al. (2023). PD-1-CD45RA+ effector-memory CD8 T cells and CXCL10+ macrophages are associated with response to atezolizumab plus bevacizumab in advanced hepatocellular carcinoma. Nat. Commun. *14*, 7825.

73. Luecken, M.D., Büttner, M., Chaichoompu, K., Danese, A., Interlandi, M., Mueller, M.F., Strobl, D.C., Zappia, L., Dugas, M., Colomé-Tatché, M., and Theis, F.J. (2022). Benchmarking atlas-level data integration in single-cell genomics. Nat. Methods *19*, 41–50.

74. Qian, J., Olbrecht, S., Boeckx, B., Vos, H., Laoui, D., Etlioglu, E., Wauters, E., Pomella, V., Verbandt, S., Busschaert, P., et al. (2020). A pan-cancer blueprint of the heterogeneous tumor microenvironment revealed by single-cell profiling. Cell Res. *309*, 745–762.

75. van den Brink, S.C., Sage, F., Vértesy, Á., Spanjaard, B., Peterson-Maduro, J., Baron, C.S., Robin, C., and van Oudenaarden, A. (2017). Single-cell sequencing reveals dissociation-induced gene expression in tissue subpopulations. Nat. Methods *14*, 935–936.

76. Alquicira-Hernandez, J., and Powell, J.E. (2021). Nebulosa recovers single-cell gene expression signals by kernel density estimation. Bioinformatics *37*, 2485–2487.

77. Guinney, J., Dienstmann, R., Wang, X., de Reyniès, A., Schlicker, A., Soneson, C., Marisa, L., Roepman, P., Nyamundanda, G., Angelino, P., et al. (2015). The Consensus Molecular Subtypes of Colorectal Cancer. Nat. Med. *21*, 1350–1356.

CellPress
OPEN ACCESS

78. Bankhead, P., Loughrey, M.B., Fernández, J.A., Dombrowski, Y., McArt, D.G., Dunne, P.D., McQuaid, S., Gray, R.T., Murray, L.J., Coleman, H. G., et al. (2017). QuPath: Open source software for digital pathology image analysis. Sci. Rep. *7*, 16878.

79. Weigert, M., Schmidt, U., Haase, R., Sugawara, K.,, and Myers, G. (2020). Star-convex Polyhedra for 3D Object Detection and Segmentation in Microscopy. In 2020 IEEE Winter Conference on Applications of Computer Vision (WACV) (IEEE), pp. 3655–3662. https://doi.org/10.1109/WACV45572.2020.9093435.

80. Dann, E., Henderson, N.C., Teichmann, S.A., Morgan, M.D., and Marioni, J.C. (2022). Differential abundance testing on single-cell data using k-nearest neighbor graphs. Nat. Biotechnol. *40*, 245–253.

81. Hänzelmann, S., Castelo, R., and Guinney, J. (2013). GSVA: gene set variation analysis for microarray and RNA-Seq data. BMC Bioinf. *14*, 7.

82. Sturm, G., Finotello, F., Petitprez, F., Zhang, J.D., Baumbach, J., Fridman, W.H., List, M., and Aneichyk, T. (2019). Comprehensive evaluation of transcriptome-based cell-type quantification methods for immuno-oncology. Bioinformatics *35*, i436–i445.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Critical commercial assays		
Collagenase P	Sigma Aldrich	Cat# 11249002001
DNAse I	Roche	Cat# 89836
DMEM	Thermofisher Scientific	Cat# 41966029
Red blood cell lysis buffer	Roche	Cat# 11814389001
40μm Flowmi Tipstrainer	VWR	Cat# 734-5950
Fetal Bovine Serum (FBS)	Thermofisher Scientific	Cat# A5256701
DMSO	Sigma Aldrich	Cat# 472301-100ML
AO/PI Cell viability kit	Westburg	Cat# LB F23001
Chromium next GEM Single Cell 5′ kit V2	10× Genomics	Cat# 1000263
Chromium Next GEM Single Cell 5′ Library and Gel Bead Kit v1.1	10× Genomics	Cat# 1000165
Chromium Next GEM Single Cell 3' Kit v3.1	10× Genomics	Cat# 1000268
NextSeq 500/550 High Output Kit v2.5 (75 Cycles)	Illumina	Cat# 20024906
NovaSeq 6000 S2 Reagent Kit v1.5 (100 cycles)	Illumina	Cat# 20028316
HiSeq 3000/4000 PE Cluster Kit	Illumina	Cat# PE-410-1001
Deposited data		
scRNA-seq of tumor, tumor-draining lymph node, metastasis and non-malignant (adjacent) samples	This manuscript	See Table S1 for details
scRNA-seq data of tumor samples	Various public studies	See Table S4 for details
Bulk RNA-seq data of tumor samples	Various public studies	See Table S4 for details
Visium spatial transcriptomics data of tumor samples	Various public studies	See Table S4 for details
Software and algorithms		
CellRanger	10× Genomics	Version 3
Seurat	https://satijalab.org/seurat/	Version 4
miloR	https://github.com/MarioniLab/miloR	Version 1.6.0
Nebulosa (R package)	https://github.com/powellgenomicslab/Nebulosa	Version 1.8.0
Adobe Illustrator	https://adobe.com/products/illustrator	Version 28.5
Biorender	https://www.biorender.com/	N/A
Corrplot	https://github.com/taiyun/corrplot	Version 0.92
ggplot2	https://ggplot2.tidyverse.org/	Version 3.5.1
Metafor	https://github.com/cran/metafor	Version 4.4–0
Scissor	https://github.com/sunduanchen/Scissor	Version 2.0.0
QuPath	https://github.com/qupath/qupath	Version 0.4.2
Immunedeconv	https://github.com/omnideconv/immunedeconv	Version 2.1.3
LISI	https://github.com/immunogenomics/LISI	Version 1.0
GSVA	https://github.com/rcastelo/GSVA	Version 2.0.5
CellChat	https://github.com/sqjin/CellChat	Version 1.1.3
StarDist	https://github.com/stardist/stardist	N/A
Zenodo	https://zenodo.org/	https://doi.org/10.5281/zenodo.16848959
Other		
Shiny app for TME exploration	This manuscript	http://apps.lambrechtslab.sites.vib.be/PanCancer-Atlas

CellPress
OPEN ACCESS

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Patient population

The study collected 230 fresh tissue samples ("orig.ident" in Table S1) from 160 patients across 9 different cancer types: breast cancer (BC, *n* = 39), CC (*n* = 15), CRC (*n* = 21), GBM (*n* = 8), HCC (*n* = 33), HGSOC (*n* = 13), HNSCC (*n* = 33), MEL (*n* = 24) and NSCLC (*n* = 44). We obtained both early and advanced stage samples for BC (*n* = 31 and *n* = 8, respectively) and NSCLC (*n* = 36 and *n* = 8, respectively). All other tumoral samples belonged to tumors collected from (locally) advanced disease stages. For some patients, multiple samples from the same tumor or from multiple sites of the same patient were sequenced, i.e., for early NSCLC and CRC (all patients), HCC (HCC P13), MEL (MEL P3) and GBM (GBM P4 and P5). In HNSCC, cells from the same dissociation were subjected to independent scRNA-seq experiments (i.e., HNSCC P1, P10, P11, P12, P14, and P16). For all sample-level analyses, samples collected from the same tumor or resulting from different scRNA-seq data of the same sample were pooled (*n* = 177). All samples were treatment-naive tumors, except for CC and some HCC tumors. Specifically, CC samples corresponded to recurrent tumors in patients receiving radio-chemotherapy or hysterectomy followed by chemotherapy (paclitaxel in combination with carboplatin) and anti-VEGFA therapy (bevacizumab) as first-line treatment, while 38.7% (12/31) of HCC patients were previously treated for HCC, undergoing liver transplantation, locoregional treatment and/or receiving prior systemic therapy (including platinum-based chemotherapy or tyrosine kinase inhibitors). Overall, tissues were collected from either the primary (53.9%), metastasized tumor (15.2%), non-malignant (adjacent) tissue (21.7%) or tLNs (tLN; 9.1%). The detailed clinical information on patients and samples of each dataset including sample and patient ID, age category, gender, cancer and tissue type, disease stage, and 10× scRNA-seq technology are provided in Table S1. The table also reports, per sample, the total number of cells, the number of cells per cell type, and the average number of detected genes. The tissue type, number of cells, samples and patients, as well as data accession number and reference for each cancer type are provided in the Table S1. The local ethics committee at the University Hospitals Leuven approved the single-cell study for each cancer type, and all patients provided written informed consent.

METHOD DETAILS

Sample collection and processing

All samples were freshly collected and rapidly subjected to tissue dissociation to obtain single-cell suspensions on ice. Tissue samples were subjected to both mechanical and enzymatic dissociation using a scalpel and digestion medium (2 mg/mL Collagenase P (Sigma Aldrich) and 0.2 mg/mL DNAse I (Roche) in DMEM (Thermofisher scientific), respectively. Red blood cell lysis buffer (Roche) was used to remove red blood cells from the cell suspension, and cells were filtered using a 40 μm Flowmi tip strainer (VWR). The number of living cells was determined using a LUNA automated cell counter (Logo Biosystems). The whole procedure was normally completed in less than 1 h.

Single-cell RNA sequencing data acquisition and pre-processing

We performed 3′ or 5′ gene expression profiling on single-cell suspensions using the Chromium Single Cell Solution from 10× Genomics according to the manufacturer's instructions. Most included samples were processed exclusively with either 5′ or 3′ scRNA-seq (*n* = 113 and *n* = 70, respectively), although for some datasets a mix of 3′ and 5′ experiments (*n* = 47) was performed (Table S1). Experiments were performed with the aim to obtain 5,000 cells per sample. All the libraries were sequenced on Illumina NextSeq, HiSeq4000 and/or NovaSeq6000 and mapped to the GRCh38 human reference genome using CellRanger (10× Genomics).

Single-cell gene expression analysis

Raw gene expression matrices were generated using CellRanger and then analyzed using the *Seurat* (v4.0.3) R package.[69] A general quality filtering was applied to the raw count matrices, removing cells expressing <200 or >6000 genes, <400 unique molecular identifiers (UMIs) and >25% mitochondrial counts. Additional more stringent quality thresholds were applied in GBM (<600 UMIs and >20% mitochondrial counts), and in early BC and HNSCC (>15% mitochondrial counts). Furthermore, since liver cells generally have high mitochondrial RNA content, we allowed up to 50% mitochondrial counts in the HCC samples. We applied DoubletFinder (v1.0)[70] on each sample to compute a doublet score for each cell, which we later used in the downstream analysis to identity doublet clusters. After quality filtering, we obtained the expression data from 611,750 single cells with on average 1,358 genes per cell detected. Overall, we retrieved >3 billion unique transcripts. Samples were then merged per cancer type, separating early and advanced cohorts for BC and NSCLC. Gene counts were log-normalized.

Single-cell RNA sequencing clustering to determine major cell types

To construct a pan-cancer atlas of the TME at the single-cell level, we first analyzed each cancer type independently. When available (i.e., for early BC and NSCLC, CRC, HCC and HNSCC), Seurat objects with the annotation of the major cell types from previous studies were used as the starting point of the analysis. Default parameters of Seurat were used. Briefly, for the clustering of all cell types, only the 2000 most variable genes were considered (computed with the *FindVariableFeatures* function). This was followed by scaling and regression for the following confounding factors: percentage mitochondrial genes, cell cycle (S and G2M scores were calculated by the *CellCycleScoring* function in Seurat), the number of UMIs (counts) and sample ID. After regression, we applied

Principal Component Analysis (PCA) to reduce the dimensionality of the data. PCs covering the highest variance in the dataset were selected for clustering and two-dimensional visualization with Uniform Manifold Approximation and Projection (UMAP). The number of selected PCs was based on elbow plots. The only exception was HCC, which required the *Harmony*[71] algorithm to regress out sample-specific effects[72] prior to clustering. The clusters consisting of distinct cell types were identified based on the expression of canonical marker genes. Differentially expressed genes (DEG) that functionally characterized the clusters were defined by *Wilcoxon* rank-sum test using the *FindAllMarkers* function from Seurat. All major cell types were identified in one clustering step, except for mast cells and DCs, which often co-clustered with other myeloid cells. These cell types were separated from the myeloid cells by clustering at higher resolution and annotating them based on established marker genes (mast cells: *MS4A2, TSPAB, CPA3*; pDC: *LILRA4* and *CXCR3*; cDCs: *CLEC9A, XCR1, CD1C, CCR7, CCL17, CCL19*; Langerhans-like DC: *CD1A, CD207*).

Comparison of dissociation biases in single-cell RNA sequencing and bulk RNA-seq across cancer types

We used our scRNA-seq atlas to determine major cell type signatures conserved across the 5 cancer types for which in-house matched bulk RNA-seq data were available: early BC (n = 3) and NSCLC (n = 4), CRC (n = 6), HCC (n = 6) and HGSOC (n = 6). DEGs were computed using the *FindConservedMarkers* Seurat function with default parameters. We retained the top 100 genes that were DE in all cancer types, and for which the maximum Bonferroni-adjusted p-value was <0.01. To make the scRNA-seq comparable with the bulk RNA-seq data, pseudobulk aggregates per sample were computed. Signature expression values in both datasets were computed by summing the counts of the genes and log-normalizing. These signatures' expression scores were then compared between scRNA-seq and bulk RNA-seq data from the same set of matched samples. By comparing the expression of cell type signatures at both the single-cell and bulk RNA levels, we were able to uncover the differences in cellular composition enrichment between the two sequencing techniques, highlighting the dissociation biases associated with their respective protocols.

Single-cell RNA sequencing clustering to determine cell subtypes

Cells assigned to the same cell type across cancer types were pooled and the *Harmony*[71] batch effect correction algorithm was then used to correct for potential batch effects arising due to different scRNA-seq technologies (5′- and 3′-scRNA-seq). The only exception was observed with Bcells, where a stronger technical batch effect was present. To address this, we applied Seurat's canonical correlation analysis, which is known for more aggressively removing technical batch effects while preserving biological variation.[73] We computed the LISI scores using the *LISI* (v1.0)[71] R package to confirm that the technical batch effects were adequately removed in each analysis. During the subclustering, we identified confounding factors that should be regressed to reduce potential technical bias, as previously described.[74] In general, we regressed for the following confounding factors: sample ID, number of UMIs (counts), percentage of mitochondrial genes, cell cycle, cancer type, IFN response score (calculated by the *AddModuleScore* function in Seurat using the gene set BROWNE IFN RESPONSIVE GENES from the Molecular Signatures Database or MSigDB v6.2) and stress score.[75] Regressing for cell cycle, IFN response and stress scores was also needed, because failure to regress for them revealed proliferating, IFN-high and sample dissociation-induced stress subtypes, respectively.[11,74] We additionally regressed out for the hypoxia signature[74] when subclustering macrophages/monocytes and Bcells to avoid the formation of subtypes driven by hypoxia. In macrophage/monocytes, regressing for stress was not needed, while in the T cell and Bcell clustering analyses we found clusters driven by the expression of hemoglobin genes, which are common contaminants from ambient RNA. These genes were excluded from the variable features before calculating PCA embeddings. Similarly, B cell receptor genes (*IGLVs, IGKVs, IGHVs*) were excluded when subclustering Bcells to avoid somatic hypermutation associated variances. Low quality clusters were identified based on low number of UMIs and genes per cell, in addition of having high mitochondrial RNA content. Doublet clusters expressed marker genes from other cell lineages and had a higher-than-expected doublet rate (>0.8% per 1000 recovered cells in the sample according to User Guide from 10× Genomics), as predicted by *DoubletFinder* (v1.0).[70] For the T_{EX} and T_{REG} subclustering analyses, we followed the same workflow as for the global T cell analysis, including sample integration with *Harmony* to correct for differences in sequencing chemistry (3′ vs. 5′) and regression of major confounding factors (sample ID, number of UMIs, mitochondrial gene percentage, cell cycle, cancer type, IFN response score), while stress score regression was not necessary. No clusters were driven by hemoglobin gene expression, and removal of hemoglobin-expressing cells was therefore not required.

Curated functional gene signatures to characterize subtypes

We curated functional signatures through an extensive literature review (Table S2) and assessed their expression across subtypes using the Seurat function *AddModuleScore* per cell. We then calculated the mean expression of each gene signature across cells belonging to individual subtypes. To enable meaningful comparisons between subtypes, we scaled the expression data across cell types (column-wise). The heatmaps were generated using the Seurat function *pheatmap*, incorporating hierarchical clustering with the default settings, which use Euclidean distance as similarity metric and complete linkage as clustering method to group signatures based on their expression patterns across different cell types. Signature scores were normalized, scaled and were then visualized using Nebulosa-generated density plots (v1.8.0)[76] using the kernel density estimation method "weighted kernel-density estimates (wKDE)."

Correlations between cell (sub)type abundances and T cell reactivity

To assess the correlation of tumor-reactive Tcells and other cell (sub)type abundances in the scRNA-seq data, we determined a 'T cell reactivity' signature consisting of the following marker genes: *ITGAE, PDCD1, CTLA4, LAG3, GZMB, PRF1, TNFRSF9*, and

TNFRSF18. The signature score for each cell was computed using the *AddModuleScore* function in Seurat, and the mean signature score across all Tcells (excluding NK cells) within each sample was defined as the 'T cell reactivity score'. In all analyses involving the T cell reactivity score, we included only tumoral samples with more than 20 Tcells (n = 118). Cell type abundances were determined per sample by dividing the count of cells belonging to a specific cell type by the total number of cells of that sample. For each subtype, the cell counts were divided by the total number of cells in the corresponding cell type. For instance, the abundance of $CD4^+$ T_{FH}-cells was calculated by dividing the count of $CD4^+$ T_{FH}-cells by the total number of T/NK cells per sample. To minimize the impact of outliers in rare cell types, we included only samples containing more than 500 cells (n = 160 when considering all samples, n = 125 when considering only tumoral samples) when analyzing cell type abundances. This threshold corresponds to the 10th percentile of the sample size distribution, a commonly used reference point in data distribution ensuring a balance between data retention and statistical robustness. Additionally, for analyses involving subtype abundances, we included only the samples containing at least 20 Tcells and macrophages/monocytes, or at least 10 Bcells, DCs, ECs, $CD8^+$ T_{EX}-cells or T_{REG}'s.

Predicting cell-to-cell interactions in single-cell RNA sequencing data

The CellChat (v1.1.3) algorithm[35] was used to predict cell-cell interactions between cell types in scRNA-seq data, using default parameters with the following exceptions: 10,000 permutations were used and cell-cell interactions between cell subtypes were not considered for groups with less than 10 cells. We focused on significant cell-cell interactions between subtypes in TLS and type-1 immunity hubs, that were analyzed separately (p-value < 0.01).

Visium spatial transcriptomics data analysis

Visium spatial transcriptomics data and the corresponding H&E-stained sections were retrieved from a total of 60 samples from the following cancer types (Table S4): BC[37] (n = 6), CRC[38,39] (n = 12), gastric cancer (GC, n = 9),[41] HCC[40] (n = 7), MEL[43] (n = 6) and NSCLC[42] (n = 20). Among BC samples, four were TNBC samples and two were ER^+ BC samples. Among CRC samples, four were classified according to consensus molecular subtypes[77] with distinct features. GC samples comprised one microsatellite instability-high (MSI-H), three microsatellite stable/EBV-associated (MSS/EBV), and five microsatellite stable (MSS) samples. HCC samples included four responders and three non-responders to neoadjuvant cabozantinib (tyrosine kinase inhibitor) and nivolumab (anti-PD1) therapy. MEL samples were categorized by tumor-infiltrating lymphocyte levels: one "brisk" (hot), three "non-brisk," and two "absent" (cold). NSCLC samples included 12 lung adenocarcinoma, six lung squamous cell carcinoma (LUSC), and two undetermined cases. Subtype-specific signatures were derived from our atlas using the *FindAllMarkers* Seurat function across pan-cancer subtypes. Marker genes were identified based on the following criteria: minpct = 0.10, min.diff.pct = 0.25, logfc.threshold = 0.25, return.threshold = 0.01, only.pos = FALSE. Among these, the top 10 genes were prioritized based on decreasing logFC and increasing adjusted p-value (Table S3). While all other subtypes could be identified, the mature and immature IgG and IgA cells had very similar expression profiles and thus were identified as a unique subgroup called IgG and IgA cells. TLS hub and type-1 immunity hub signatures were derived from pooling the signatures from each subtype belonging to the corresponding hub. All Visium samples were merged using the *merge* Seurat function and the expression levels of genes were log-normalized. Both subtype-specific and hub-specific signatures were then computed for each spot using the *AddModuleScore* Seurat function. Pearson correlations between each pair of cell subtypes within hubs were computed across spots and combined into a correlation heatmap with the *corrplot* (v0.92) R package. The bar plots were generated by *ggplot2* (v3.5.1) R package to compare the average hub scores of each sample across cancer types.

Akoya spatial proteomics data analysis

We analyzed single-cell spatial protein profiling data from a 10 μm thick FFPE tissue section of an HNSCC sample, obtained from patient 14 of the window-of-opportunity phase I/II DUTROLASCO study (NCT03784066). This dataset was previously generated[44] using the PhenoCycler Fusion platform (Akoya Biosciences) with a panel of 60 immuno-oncology markers. The images were analyzed in QuPath (v0.4.2),[78] with the *StarDist* extension[79] used for cell segmentation, as previously described.[44] We exported a data matrix containing antibody intensities (mean of pixel intensity) per cell along with x and y spatial coordinates from QuPath into Seurat for analysis. Markers for subtype-specific and hub-specific signatures were chosen based on the Visium spatial transcriptomics analysis (Table S3), with further selection according to antibody availability. Then, markers uniquely present in either TLS or type-1 immunity hub were selected. The TLS-specific score was determined using the following markers: CD20, CD21, CD79a, CX3CR1, and IDO1. The type-1 immunity score was defined based on CD66, CD8, FOXP3, GZMB, and LAG3 expression.

Hub-specific intensity scores were computed for each cell using Seurat's *AddModuleScore* function. Plots illustrating the TLS and type-1 immunity hub scores were generated with the blending option of *ImageFeaturePlot* function in Seurat. Then, we selected a 0.25 mm^2 hub-enriched area for each hub, and within this area we identified hub-specific subtypes combining different antibodies. The hub-specific subtypes and corresponding set of antibodies used to identify them at the spatial proteomics level are listed below: GC B cells (CD79a, CD20), plasma cells (CD79a, CD138), B_{REG}'s (CD79a, HLA-DR, PD-1, GMZB), $CD4^+$ T_{FH}-cells (CD3e, CD4, PD-1), MigDCs (CD68, CD11c, PD-L1), Inflammatory Mac (CD68, HLA-DR, PD-L1, IDO1), ECs (CD31, CD34), $CD4^+$ T_{REG}'s (CD3e, CD4, FOXP3), $CD8^+$ T_{EX}-cells (CD3e, CD8, PD-1, IFNG), $CD4^+$ T_{H1}-cells (CD3e, CD4, PD-1, Tbx21), proliferating Tcells (CD3e, CD4, Ki67), and neutrophils (MPO).

Public single-cell RNA sequencing datasets collection

We created a cohort of publicly available scRNA-seq data starting from either raw or processed data (Table S4). The analysis was initiated from the raw read count matrices. If this was not available, the raw sequencing reads were processed to obtain the count matrices, as previously described for our own data. We used all available datasets generated using 10× Genomics technology, while selecting only datasets with no enrichment for specific cell types during tissue dissociation. Overall, we retrieved studies from the following cancer types: breast cancer (n = 4 studies), CRC (3 studies),[5,48,49] HCC (5 studies)[4,50–53] and NSCLC (5 studies).[54–58] Analyses focused exclusively on samples with a minimum of 500 cells. Overall, these public datasets generated a large amount of single-cell data: from BC (n = 469,008), CRC (n = 240,218), HCC (n = 321,592) and NSCLC (n = 289,327), resulting in a total of 1,320,145 cells from 320 patient samples (BC = 104, CRC = 99, HCC = 47, NSCLC = 70). For each cancer type, major cell types (Tcells, Bcells, macrophages/monocytes, and DCs) were identified by clustering and manual annotation with canonical marker genes, while within each cell type, subtypes were identified by label transfer. Particularly, we applied the reference-based integration method of Seurat[69] to annotate subtypes of Tcells, Bcells, macrophages/monocytes, and DCs using our pan-cancer objects as a reference. This allowed us to identify all the previously identified subtypes, except for mRegDCs and mQuiescDCs, which expressed similar expression profiles and thus were combined into a group called migratory DC (MigDC). For the gene expression heatmaps, counts were log-normalized and a standardized Z score was computed over the cell types in each heatmap.

T cell expansion analysis

We leveraged two scTCRseq datasets obtained from patients for which scRNA-seq was already included in our atlas. Paired samples in these patients were collected pre-treatment, and on-treatment after 1 dose of ICB. Included studies involved early BC[11] patients (n = 26) and HNSCC[44] patients (n = 15) undergoing neoadjuvant immunotherapy with pembrolizumab (anti-PD1) or durvalumab (anti-PD-L1) with or without tremelimumab (anti-CTLA4), respectively. We measured T cell expansion as the number of expanded T cell clonotypes identified by scTCRseq data comparing on-versus pre-treatment tumors in both studies. ‘‘Expanders’’ (Es; BC, n = 9; HNSCC, n = 7) showed significant T cell clonal expansion (>30 shared TCRs) after ICB treatment, while ‘‘non-expanders’’ (NEs; BC, n = 17; HNSCC, n = 8) showed little to no T cell clonal expansion. We considered T cell expansion as an early surrogate marker of clinical response to ICB. To identify specific subtypes associated with expansion, we performed differential abundance analysis between cell subtypes from Es and NEs (combining both BC and HNSCC) using a tool for differential abundance testing on K-nearest neighbor graph neighborhoods, implemented in the R package *miloR* (v1.6.0).[80] For graph building, d and k parameters were defined selecting the same number of PCs for the corresponding cell type analysis, as recommended.

Collection of public bulk RNA-seq datasets with immune checkpoint blockade response

We also examined the association between cell subtype abundances and clinical outcome defined as PFS in 4 clinical trials involving ICB with the anti-PD-L1 inhibitor atezolizumab (Table S4). Particularly, we retrieved read counts data from i) OAK (https://clinicaltrials.gov/study/NCT02008227), a Phase III, Open-Label, Multicenter, Randomized Study to Investigate the Efficacy and Safety of Atezolizumab (Anti-PD-L1 Antibody) Compared With Docetaxel in Patients With NSCLC After Failure With Platinum Containing Chemotherapy; ii) POPLAR (https://clinicaltrials.gov/study/NCT01903993), A Randomized Phase 2 Study of Atezolizumab (an Engineered Anti-PD-L1 Antibody) Compared With Docetaxel in Participants With Locally Advanced or Metastatic NSCLC Who Have Failed Platinum Therapy; iii) IMvigor210 (https://clinicaltrials.gov/study/NCT02951767), A Phase II, Multicenter, Single-Arm Study of Atezolizumab in Patients With Locally Advanced or Metastatic Urothelial Bladder Cancer; iv) GO30140 (https://clinicaltrials.gov/study/NCT02715531), An Open-Label, Multicenter Phase Ib Study of The Safety and Efficacy of Atezolizumab (Anti-PD-L1 Antibody) Administered in Combination With Bevacizumab and/or Other Treatments in Patients With Solid Tumors, and IMbrave150 (https://clinicaltrials.gov/study/NCT03434379), A Phase III, Open-Label, Randomized Study of Atezolizumab in Combination With Bevacizumab Compared With Sorafenib in Patients With Untreated Locally Advanced or Metastatic HCC. Altogether, we curated a dataset comprising 937 ICB-naive samples based on these clinical trials.

Scissor analysis

Scissor[64] was used to integrate bulk publicly available RNA-seq data and patient survival data with our pan-cancer scRNA-seq data to identify individual transcriptomes associated with survival in pre-treatment tumors. PFS and bulk RNA-seq data were obtained from the aforementioned studies. Briefly, the Scissor algorithm first quantifies the similarity between single-cell data and bulk data by computing the correlation between each single-cell transcriptome and each bulk RNA-seq sample. Then, Scissor optimizes a regression model between the correlation matrix and PFS. Specifically, a built-in Cox regression model was used to identify cells associated with PFS. This analysis was performed separately for each bulk RNA-seq dataset and major cell type. Alpha values were chosen from 0.0625, 0.01, 0.001, and 0.0001, depending on the number of samples and cells in each analysis. Output cells were classified into i) Scissor-positive cells (associated with longer PFS), ii) Scissor-negative cells (associated with shorter PFS) and iii) cells showing no clear correlation with any patient sample (these were excluded from further analysis). Pooling the results from individual cells enabled us to unbiasedly identify which cell subtypes were linked to either longer or shorter PFS across clinical studies. To achieve this result, we performed an enrichment analysis to calculate the enrichment of cells associated with longer PFS in each subtype, using log-transformed odds ratios derived with the *Metafor*[59] R package in each individual clinical study. We then plotted the average odds ratios across studies.

CellPress
OPEN ACCESS

The Cancer Genome Atlas analysis

We retrieved bulk RNA-seq data and clinical metadata of 14 TCGA datasets, comprising a total of 7,493 samples (BLCA, *n* = 411; BRCA, *n* = 1,084; COAD, *n* = 594; HNSC, *n* = 523; KIRC, *n* = 512; LGG, *n* = 514; LIHC, *n* = 372; LUSC, *n* = 487; OV, *n* = 585; PRAD, *n* = 494; SKCM, *n* = 448; STAD, *n* = 440; THCA, *n* = 500; UCEC, *n* = 529). We conducted two types of analyses on the TCGA datasets. First, to evaluate the distribution of TLS and type-1 immunity hubs across cancer types, we calculated hub-specific signature scores (Table S3) using gene set variation analysis (GSVA) using the *GSVA* R package (v2.0.5).[81] Mean and median hub scores were computed and visualized across cancer types. Second, we evaluated the associations between TME hub scores and clinical features. Immune cell infiltration (ImmCellInf) was estimated using the *immunedeconv* R package (v2.1.3),[82] which applies the ESTIMATE algorithm to calculate scores for immune, stromal, and tumor components, as well as tumor purity. Pearson correlation analysis was performed to examine the relationships between TME hub scores and clinical variables, including ImmCellInf, total mutation count, age at diagnosis, disease stage, tumor break load, aneuploidy score, and fraction of genome altered. Notably, when correlating hub-specific signatures (Table S3) with the expression of selected cell type profiles, we used $CD8^+$ T_{EX}-cell (*CD3E, CD8A, GZMB, CXCL13, GNLY, CCL5, CCL4L2, CCL4, GZMA, NKG7, LAG3*) gene signature derived from our scRNA-seq dataset.

QUANTIFICATION AND STATISTICAL ANALYSIS

To compare cell type or subtype abundances across different cancer types, we used the *Wilcoxon* statistical test with Holm correction to determine adjusted *p*-values. The top three adjusted *p*-values are shown. Spearman correlation analysis was performed on the abundances of cell (sub)types, as well as between the abundances of cell (sub)types and signature expression. Unsupervised hierarchical clustering was applied to the rows and/or columns using the complete linkage method and Euclidean distance. Combined cell frequencies of both hubs were calculated as the sum of cells from the cell subtypes belonging to the hub divided by the total amount of cells in each sample (excluding doublets and low-quality cells). A Student's *t*-test was applied when calculating the fraction of Scissor-positive or Scissor-negative cells. In the latter analysis, we excluded results with fewer than 10 Scissor-identified cells per patient. We conducted the analysis by either stratifying samples based on clinical trials across the pan-cancer scRNA-seq dataset or by stratifying for the cancer types analyzed in our scRNA-seq cohort. Boxplots display the lower quartile, the median (center line), the upper quartile, as well as the mean (dot).